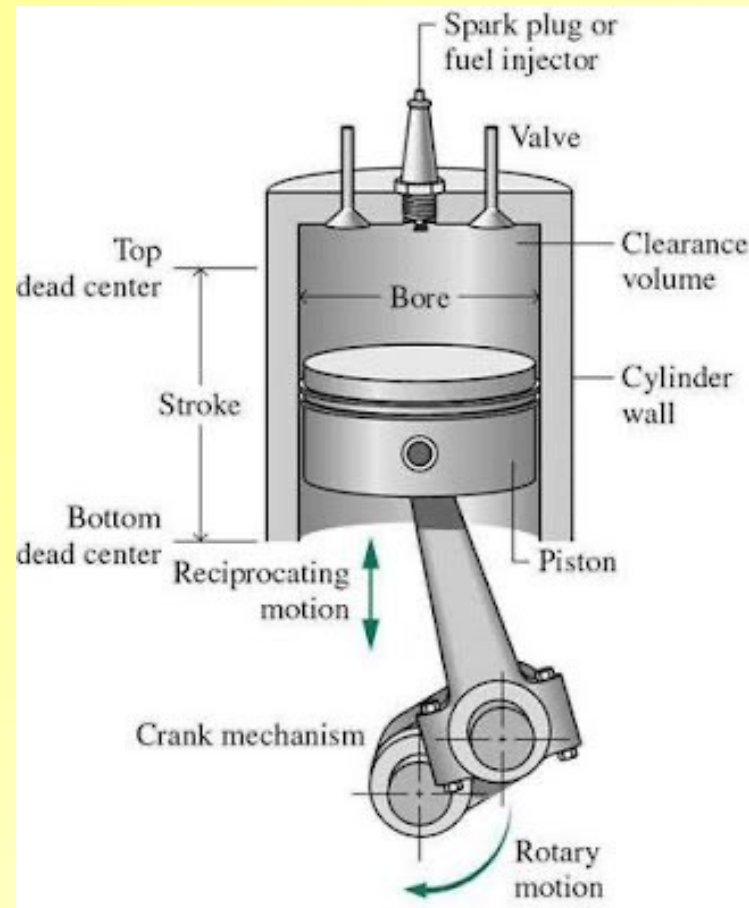




INTERNAL COMBUSTION ENGINES



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SYLLABUS



S.I. Engines: Deviation from the ideal cycle, combustion losses, flame development and propagation, effects of operating variable on detonation; Detonation rating of engine fuels, HUCR Dope, preignition combustion chambers, Mechanics of carburetion, MPFI systems, mixture requirement and combustion.

C.I. Engines: Stages of combustion, Diesel knock, effect of operating variables, knock rating of C.I. engine fuels, Cetane number; Effect of additives, combustion chambers. Requirements of fuel injection system and its mechanism, Fuel filters.

Lubrication and Cooling: Forced, splash, by-pass and direct, Blow by and crankcase dilution.

Air and water cooling, Thermostatic control.

Performance and Testing: High speed indicators, various efficiencies, Morse Test and Willan's line; torque and mean effective pressure, performance and heat balance sheet, Measurement of volumetric efficiency, Effect of atmospheric condition on performance of I.C. Engines; Supercharging; high altitude problem of I.C. Engines, methods and types of supercharging; suitability of S.I. and C.I. Engines;

Air Pollution: Vehicular emission, its norms, Causes and control of air pollution from diesel and petrol engines, catalytic converter.

Introduction to dual fuel and multi-fuel engines; free-piston engine and rotary combustion engine.



INTRODUCTION



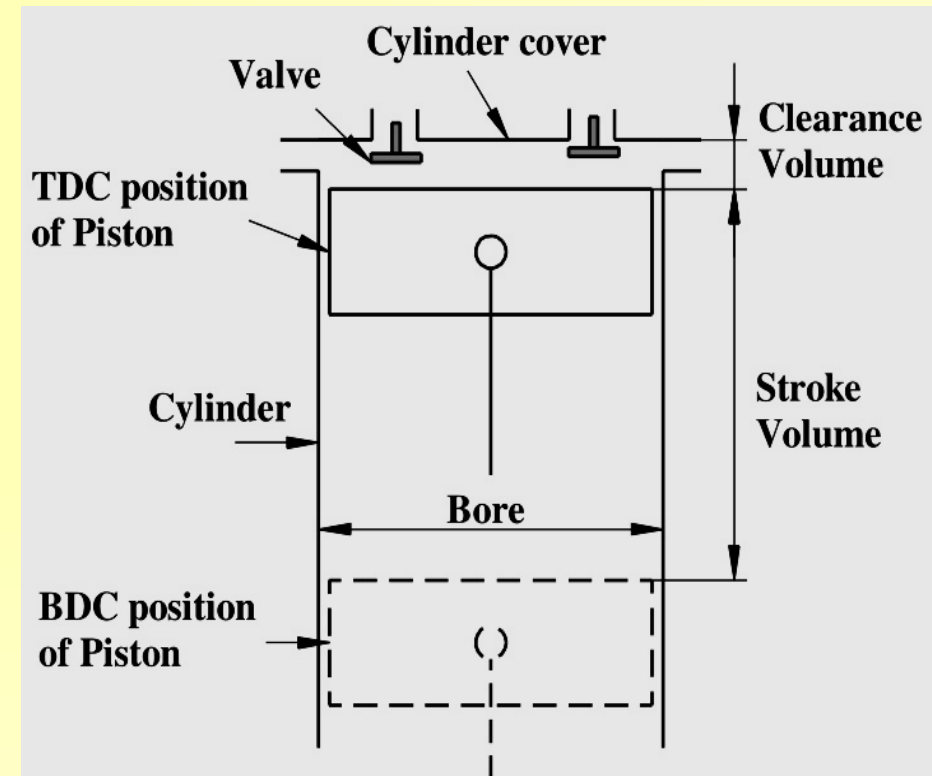
- *Heat engine: It is defined as a system that converts thermal energy into mechanical work, in which*
 - ✓ Working fluid is '**Heat**',
 - ✓ First chemical energy of fuel is converted into '**Heat Energy**' and
 - ✓ This heat energy is further convert into mechanical energy to do work.
- *The heat engines can be classified into two groups;*
 - ✓ 1. Internal combustion engines:
 - *If combustion of fuel takes place inside the engine cylinder and heat is further used into mechanical energy/ work done are known as I.C. engines.*
 - *Examples: scooter, car, bus, ship, train etc. engines.*
 - *In 1876 four stroke engine based on Otto cycle was developed by a German engineer, Nicholas Otto.*
 - *Diesel engine was was developed by another German engineer Rudolf Diesel in year 1892.*
 - ✓ 2. External combustion engines:
 - *if the combustion of fuel takes place outside the combustion chamber of heat engine and combustion products does not come in contact of engine cylinder are known external combustion engines.*
 - *The working is further taken to a machine through pipeline*
 - *This heat energy is here converted into mechanical work is known as external combustion engines.*
 - *Examples: steam engines, steam turbines, gas turbines etc.*
- *S I Engine: an engine in which combustion process in each cycle is started by use of a spark plug.*
- *C I Engine: an engine in which combustion process is started by self ignition of air-fuel mixture causing high temperature generation in combustion chamber due to high compression ratio.*



I C ENGINE TERMINOLOGY

➤ Terminology

- ✓ Bore: internal diameter of engine cylinder
- ✓ Stroke: distance $L = 2R$ between TDC & BDC,
- ✓ TDC/IDC: top dead point, $v_p = 0$, motion turns
- ✓ BDC/ODC: bottom dead point, $v_p = 0$, motion turns
- ✓ Swept vol: vol occupied between TDC & BDC
- ✓ Clearance vol: vol between TDC & cylinder head
- ✓ Clearance ratio: $CR = \frac{V_c}{V_s}$
- ✓ Total volume: sum of swept & clearance volume
- ✓ Compression ratio: total cy vol to V_c , $CR = \frac{V_T}{V_c}$
- ✓ A/F mixture: Otto cycle $\sim 15:1$,
Gas Turbines $\sim 50:$ to $250:1$
- ✓ Piston speed, $N = \frac{2LN}{60}$. rpm





AIR STANDARD OTTO-CYCLE

➤ *Air standard Otto cycle is an ideal cycle that approximates a spark ignition IC engine.*

➤ *Modelling of air standard cycles*

- ✓ Working fluid is homogeneous and follows Gas Laws, $pV = mRT$
- ✓ All processes are internally reversible specific heat of air does not vary with temp
- ✓ Compression replaced by heat input and exhaust by heat rejection

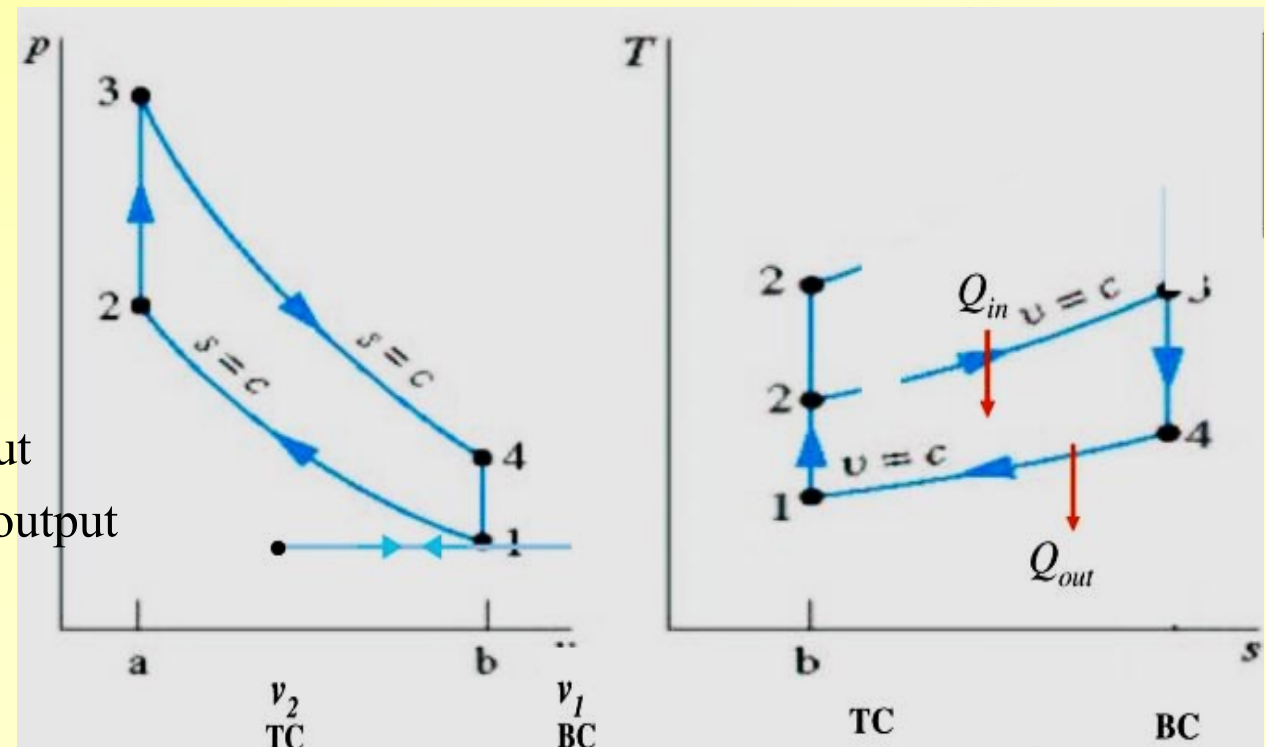
➤ *Consist four internally reversible process*

- ✓ 1-2 isentropic compression
- ✓ 2-3 iso-cohric heat-addition
- ✓ 3-4 isentropic expansion
- ✓ 4-1 iso-cohric heat-rejection

➤ *Otto Cycle Working*

- ✓ Area under curve 1-2-a-b-1, represent compression work input
- ✓ Area under 3-4-b-a-3, is work output
- ✓ Enclosed area 1-2-3-4-1, net work output
- ✓ Efficiency of Otto cycle,

$$\eta_{\text{otto}} = 1 - \frac{1}{r^{\frac{1}{\gamma}}}$$





AIR STANDARD DIESEL-CYCLE

➤ *Air standard Diesel cycle that approximates a compression ignition IC engine, consists of,*

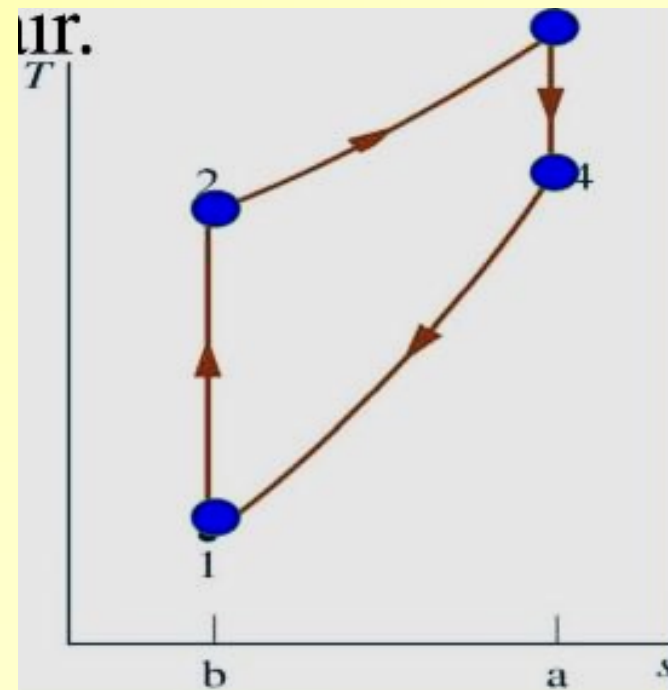
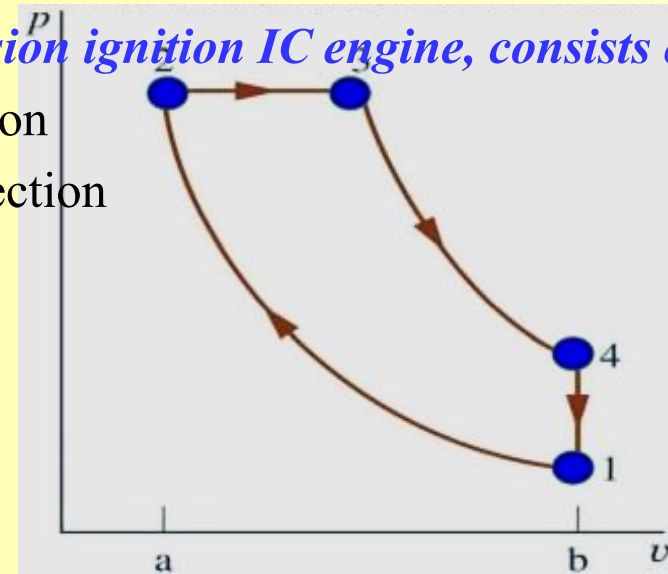
- ✓ 1-2 isentropic compression, 2-3 iso-baric heat-addition
- ✓ 3-4 isentropic expansion and 4-1 iso-choric heat-rejection

➤ *Diesel Cycle Working*

- ✓ Area under curve 1-2-a-b-1 is W_{in}
- ✓ Heat supplied is $\int Tds$.
- ✓ Area 2-3-4-b-a-2 is W_{out}
- ✓ Enclosed area 1-2-3-4-1 is $W_{net} = \int p dv$
- ✓ Diesel $\eta_{diesel} = 1 - \frac{1}{(r)^{\gamma-1}} \left[\frac{\rho^{\gamma} - 1}{\gamma(\rho - 1)} \right]$

➤ *Power Generation Air Standard Cycle Parameters*

- ✓ ASE $\eta_{th} = \frac{W.D.}{Heat\ supplied} = \frac{Q_s - Q_r}{Q_s}$
- ✓ Relative efficiency, $\eta_r = \frac{\eta_{actual}}{\eta_{th, ASE}}$
- ✓ Sp work output, $W_{sp} = W_{out}/kg\ of\ air$
- ✓ Sp air consumption, $SAC = \frac{3600}{W_{net}}$
- ✓ Work ratio, $WR = \frac{W_{net}}{W_{+ve}}$





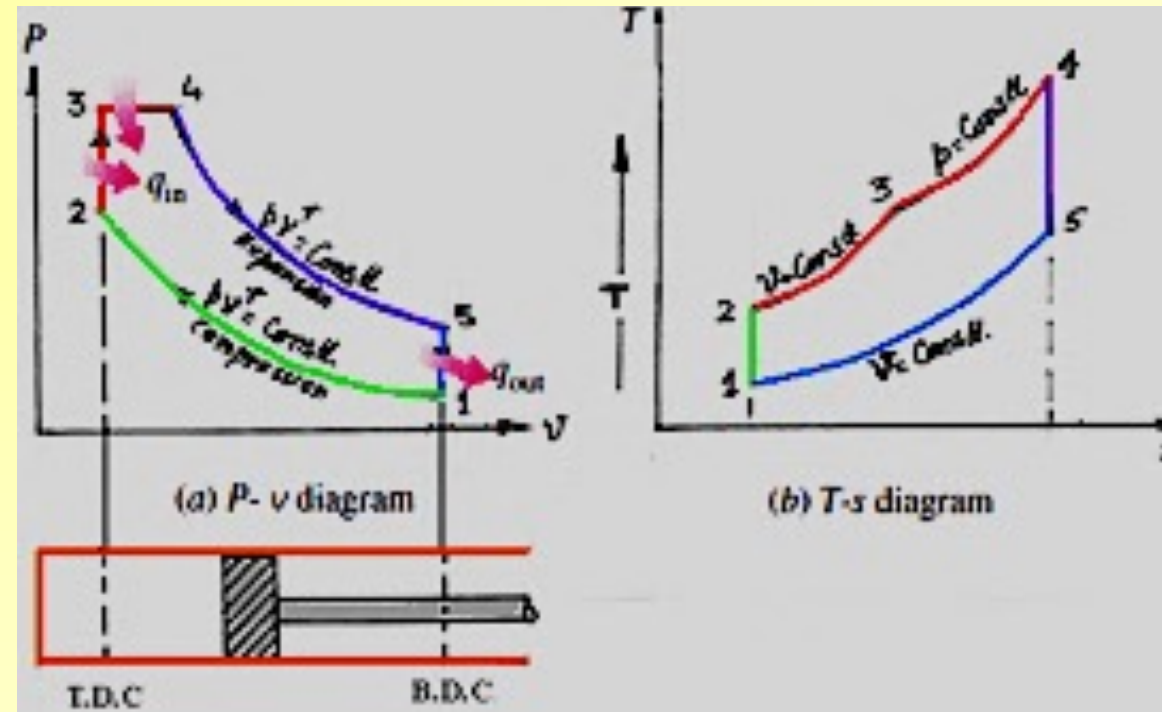
AIR STANDARD DUAL-CYCLE

- *Air standard Dual cycle is an ideal cycle that approximates an SI and CI type IC engine.*
- *It consists of five internally reversible processes in series:*

- ✓ 1-2 isentropic compression
- ✓ 2-3 iso-cohric heat-addition
- ✓ 3-4 iso-baric heat-addition
- ✓ 4-5 isentropic expansion
- ✓ 5-1 iso-cohric heat-rejection

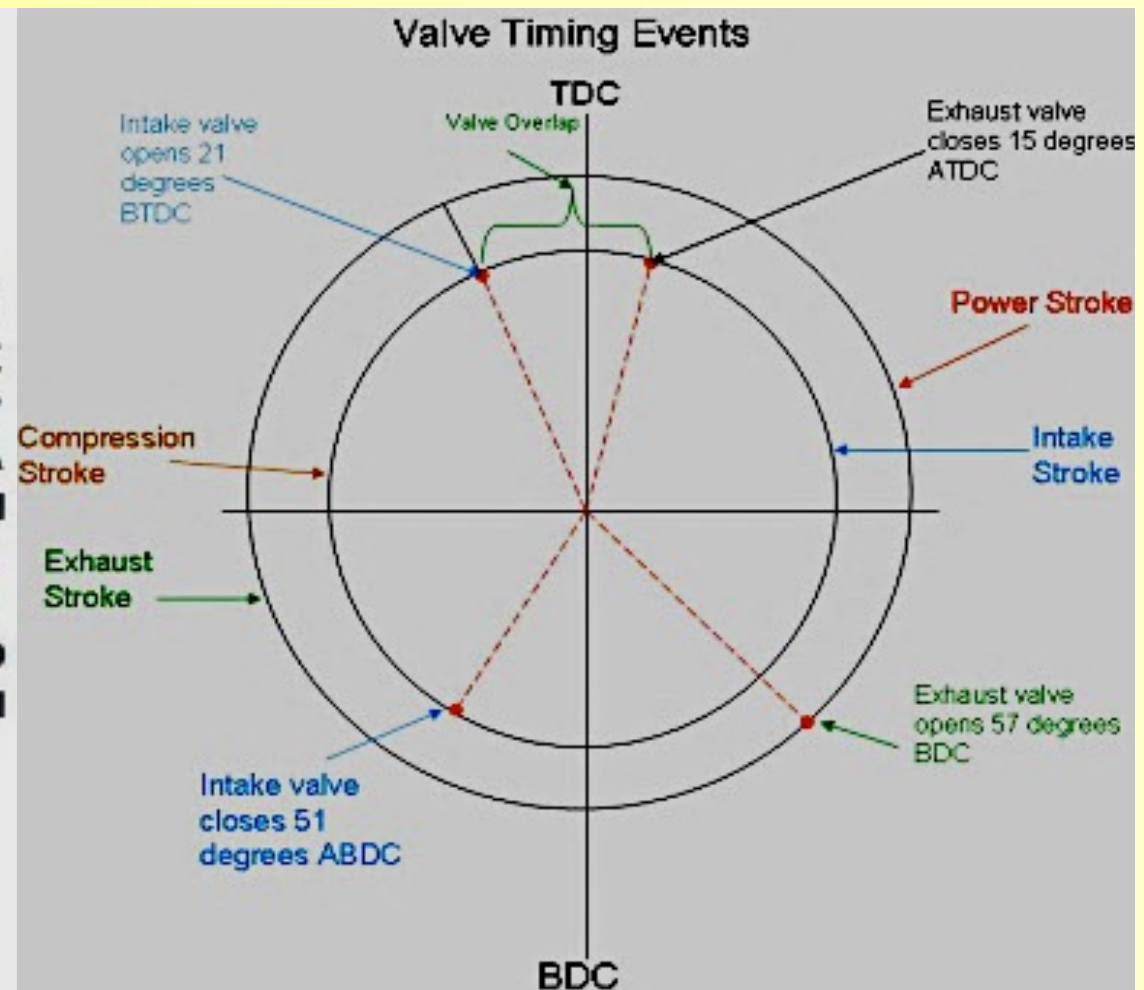
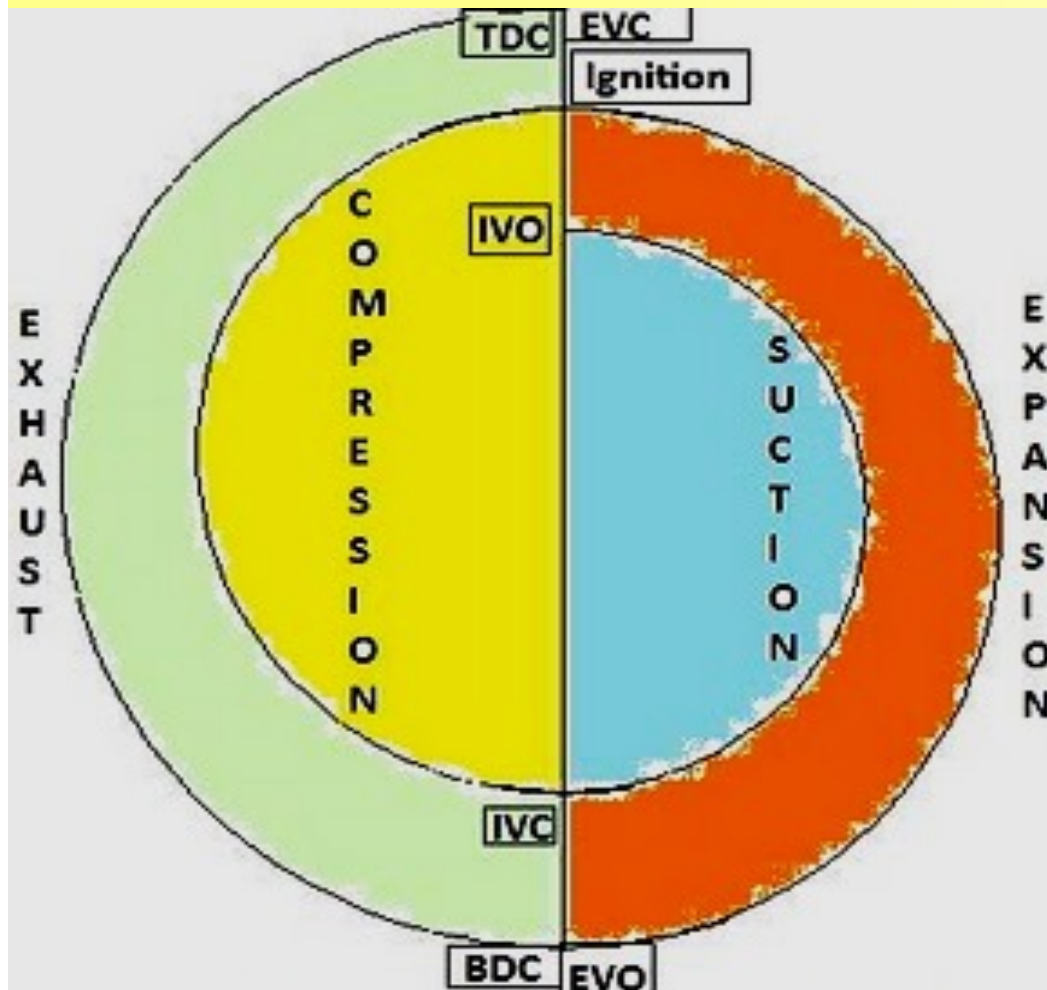
➤ Dual Cycle Working

- ✓ Area under curve 1-2-b-a-1 represent W_{in}
- ✓ Heat supplied in process 2-3 at $V=c$ &
- ✓ 3-4 at $p=c$ as $\int Tds$.
- ✓ Area under curve b-2-3-4-5-a-b represent W_{out} .
- ✓ Enclosed area 1-2-3-4-5-1 present net work output $\int pdv$
- ✓ Efficiency of Dual cycle, $\eta_{diesel} = 1 - \frac{1}{(r)^{\gamma-1}} \left[\frac{\alpha \rho^{\gamma} - 1}{(\alpha-1) + \alpha \gamma (\rho-1)} \right]$





THEORETICAL & ACTUAL VALVE TIMINGS

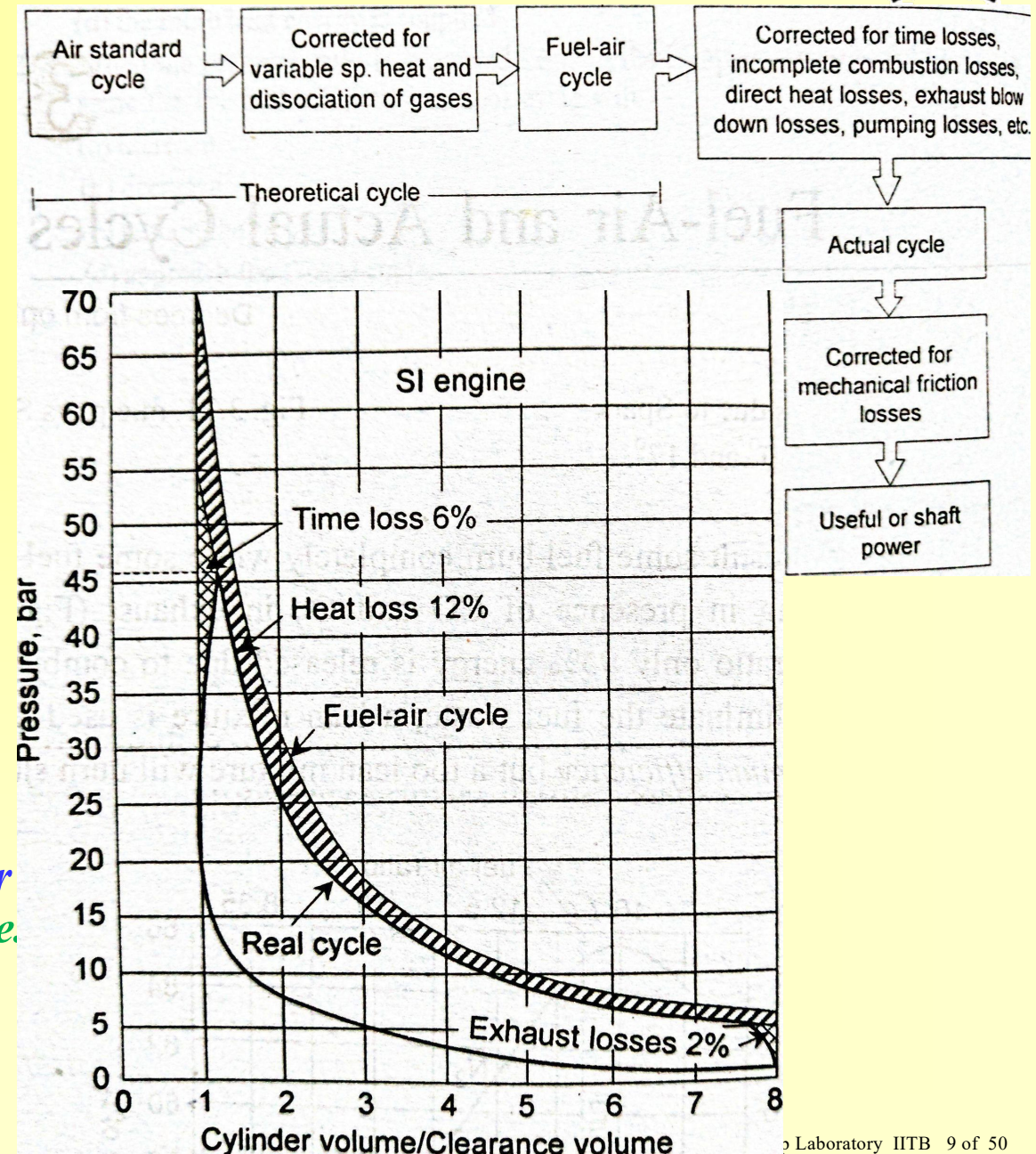




FUEL-AIR & ACTUAL CYCLES



- The η_{th} of air standard cycle is much higher than the Actual cycles, attributed to,
 - ✓ 1. Working fluid is **Real** and its **dilution**
 - ✓ 2. Losses due to change in C_p with temp
 - ✓ 3. Losses due to **Dissociation** of combustion products at high temperature
 - ✓ 4. **Time loss** in mixing of air-fuel & progressive combustion process
 - ✓ 5. **Incomplete** combustion losses
 - ✓ 6. **Heat losses** to CC walls, head, piston
 - ✓ 7. **Blow-down** losses
 - ✓ 8. **Exhaust** heat losses
 - ✓ 9. **Pumping** losses etc.
- If losses due to (1, 2, 3) are subtract from air standard cycle, it approximates **fuel-air cycle**.
- If remaining losses (4 to 9) are subtracted from Fuel-air cycle, the resulting cycle will approximate the **Actual cycle**.



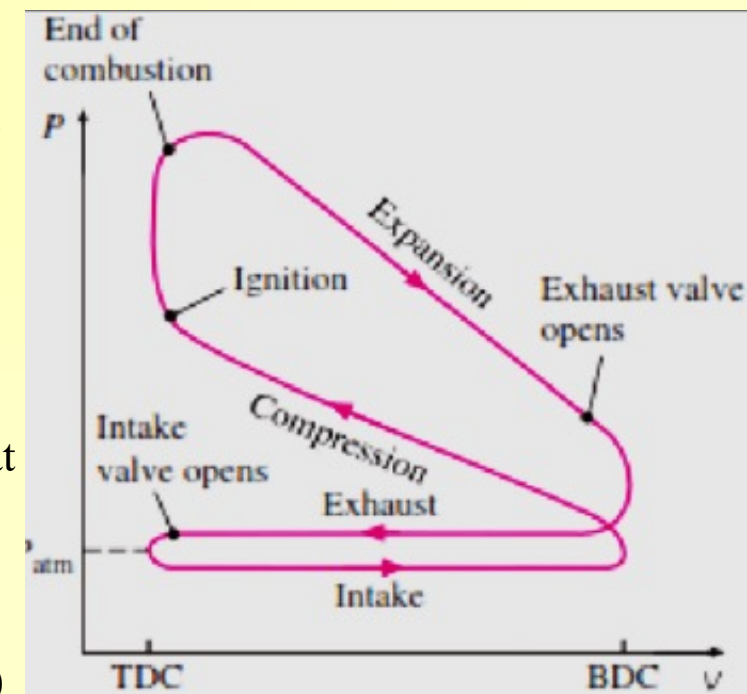
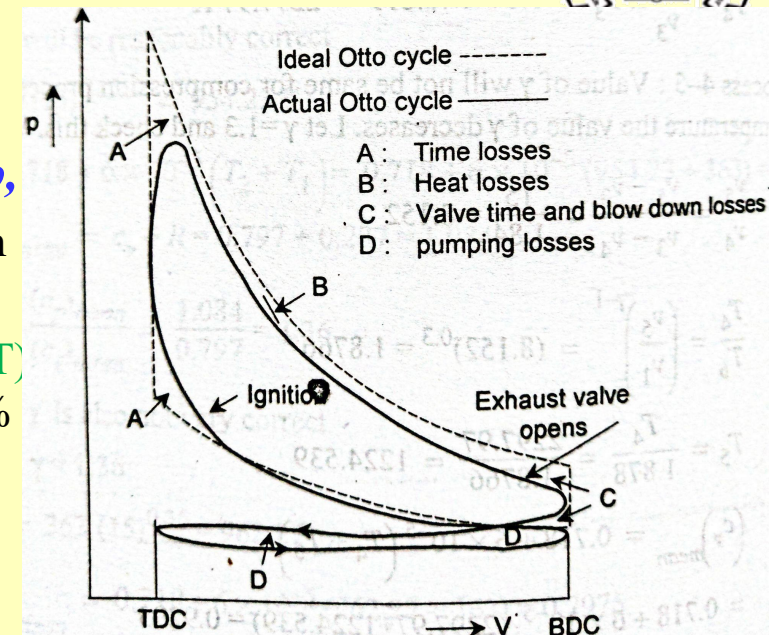


COMBUSTION LOSSES IN SI ENGINES



➤ *There is great variation between Ideal / Air standard & Actual cycle efficiencies, so $\eta_{std} = 56.5 > \eta_{act} = 30.5$. The difference is due to,*

- ✓ **Working fluid** is not air but air+fuel mixture (**real fluid**) combustion products, residual gases; so properties different than air, $pV \neq mRT$.
- ✓ **Variable sp heats**: in general c_p are not constant. It's a **function of (T)**
 $C_p = a + bT + cT^2 + dT^3 + \dots$ C_p increases with T, reduce work $\sim 12\%$
- ✓ **Dissociation**: at high-temp combustion products **break-up** into CO, H_2 & O_2 absorb heat (endo-thermic), reduce T, p, WD/power & η_{th}
- ✓ **Time loss**: combustion doesn't occur instantly at $V/p = c$, (area-A) takes **time**, depends on flame **vel** & **distance** travelled, incomplete combustion reduces WD $\sim 6\%$ & η_{th} , spark bTDC $\sim 40^\circ$ burn 90% A+F
- ✓ **Heat losses**: in ideal cycle no heat loss accounted; taken compression & expansion isentropic but due to friction **$s \neq c$** and heat transfers to cyl walls, head, coolant, surroundings, **reduces** WD $\sim 12\%$ (area-B).
- ✓ **Exhaust Dilution**: fresh charge **dilution** (mix with exhaust gases) in V_c , fresh charge surrounded by combustion products, can't burn, also reduces $\sim 2\%$ work done; so, EV should open $\sim 35-75^\circ$ bBDC.
- ✓ **Valve timing and Blow down losses**: in ideal cycle, instantaneous heat addition/rejection is assured but actually it is not possible due to **inertia** of valves & exhaust gases. Therefore EV should opens **prior time** when piston reaches bBDC, reduces the net work done (area-C).
- ✓ **Pumping loss**: work needed in intake/exhaust (2% work loss, area-D)

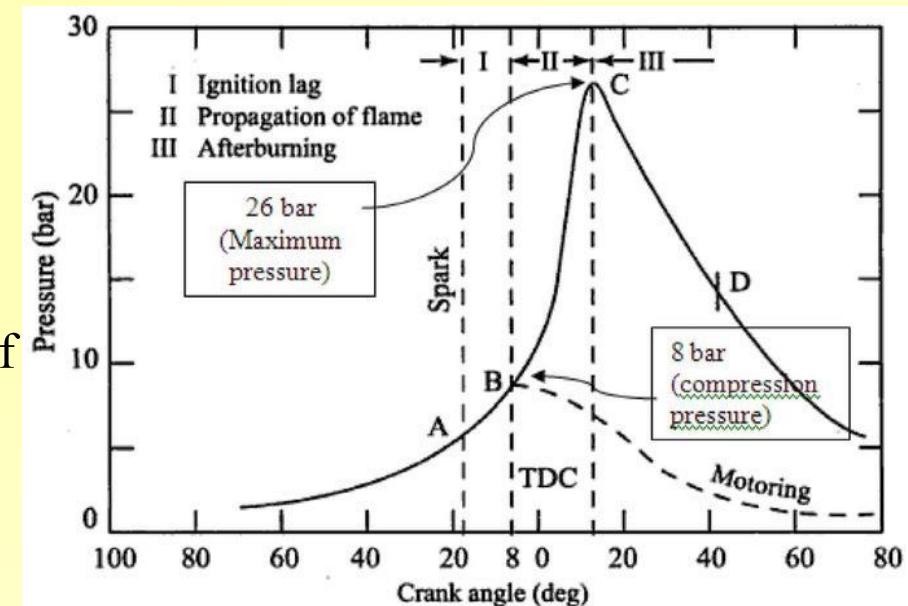




SI ENGINE : STAGES OF COMBUSTION



- ✓ A **high-temp spark**, sufficient to heat mixture given at 'A' **bTDC** during compression stroke is '**Angle of Advance**'
- ✓ Increase/decrease in set-angle is **over advanced/retardation**
- ✓ Time **interval** of mixture near spark plug get heated to self/ auto-ignition temp, rise a self-propagating nucleus of flame is **Delay period/Ignition lag** 'A-B' is very short ~ 0.002 sec
- ✓ $t_{\text{ignition delay}} = 0.08 \frac{1}{p^{1.5}(\text{MPa})} e^{\frac{3800}{T(\text{K})}} (\text{ms})$
- ✓ After **ignition delay** rise in gas temp, **Pressure-Force** in CC
- ✓ After ignition, molecule heated to self-ignition temp & **pre-flame** travel till 'B' on compⁿ curve, actual **Burning** starts
- ✓ **Propagation** of flame through unburnt mixture will follow if spark heat > reaction rate, it **overcomes** the heat losses.
- ✓ Once combustion started & reaction is well under way, the spherical **flame-front** advances & create new flame fronts.
- ✓ At ignition, combustion starts, a rapid rise in pr & temp cause **peak pr** at 'C' & flame propagate along curve **B-C**.
- ✓ Peak pr at 'C' If combustion process continues after 'C' during expansion stroke, called **After burning** 'C-D'.
- ✓ Spark adjusted to produce high **peak pr-force** close to TDC



Stages of Combustion in an SI Engine

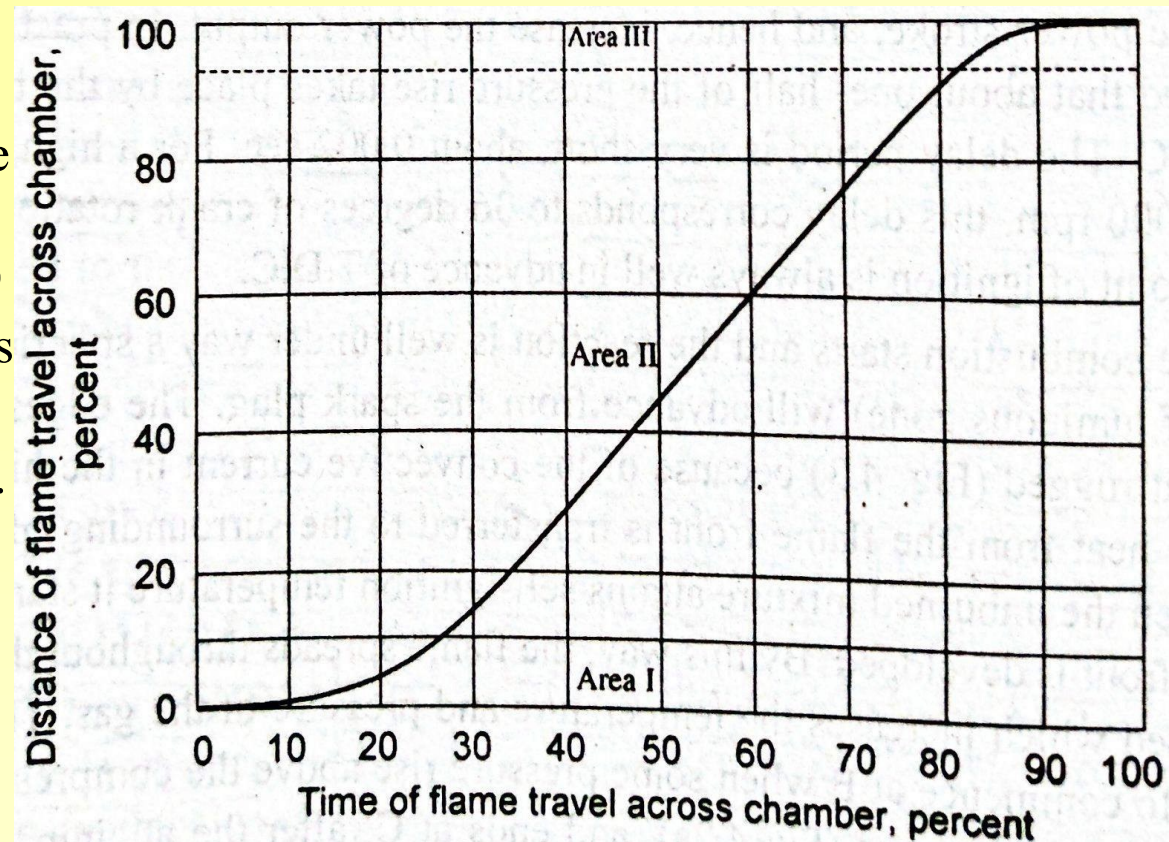


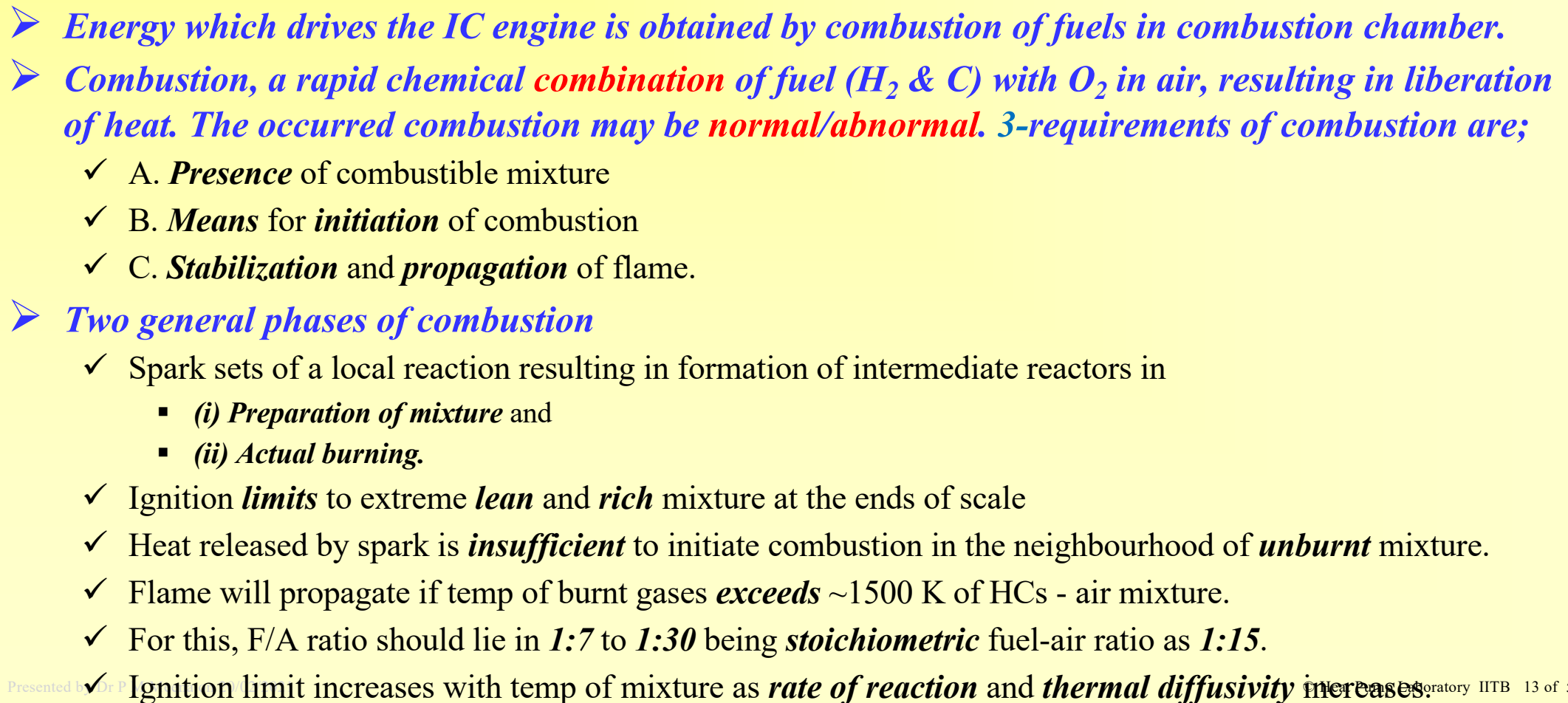
COMBUSTION ZONES : SI ENGINES



➤ Flame pass through 3-distinct stages.

- ✓ Initially the flame front progresses at low velocity due to “**slow transportation rate**” & “**low turbulence.**” (*Area-I*)
- ✓ As the flame front **proceeds** into more turbulent zone, commences and consume more mass of mixture, flame progress is more **rapid** but at constant rate (*Area-II*)
- ✓ The **average velocity** of flame during this travel period is **flame velocity** (*Area-II*). Flame travel is fastest in **richest** mixture.
- ✓ At **end** of flame travel the volume of turbulent **charge** is reduced and the transportation rate reduces to **negligible** value, which reduces the speed of flame propagation, (*Area-III*).
- ✓ Moreover the flame enters a zone of relatively **low turbulence** which in turn **reduces** the combustion reaction rate.







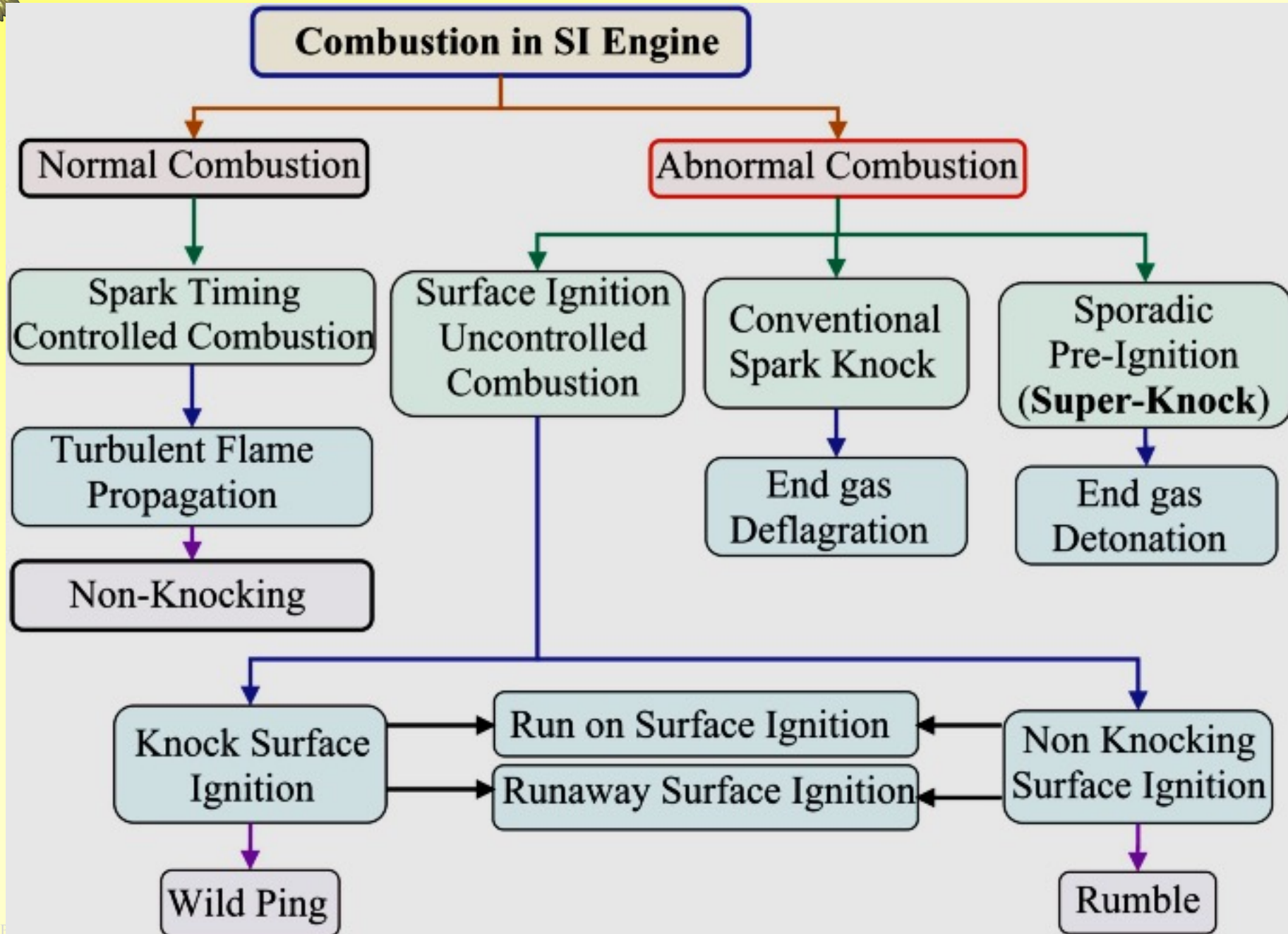
COMBUSTION REQUIREMENTS

➤ *The effect of engine operating variables on pr and temp within the cylinder are:*

- ✓ Compression Ratio:
 - *Fuel-air efficiency increases with increase in T , p & compression ratio*
- ✓ Fuel-Air Ratio:
 - *Power output becomes maximum at stoichiometric mixture strength then decreases*
- ✓ Efficiency:
 - *Decreases with lean mixture due to lower combustion temperature also decreases the exhaust temperature, dissociation and specific heat so efficiency is higher and approaches to the air standard cycle.*
- ✓ Maximum Power:
 - *Fuel-air ratio affects the maximum power output of the engine, encourage each with a richer mixture but decreases after reaching the stoichiometry mixture.*
- ✓ Maximum Temperature:
 - *Rich mixture causes more fuel to combine with oxygen, increases the temperature and reaches a certain optimum point than slightly decreases the power output.*
- ✓ Maximum Pressure:
 - *The pressure of a gas depends on temperature and number of fuel molecules because of increasing the number of molecules becomes maximum then decreases.*
- ✓ Exhaust Temperature:
 - *Exhaust gas temperature is maximum at chemically correct mixture then decreases*
- ✓ Mean Effective Pressure:
 - *p_m increases with compression ratio, become max for slightly rich mix than chemically correct ratio then decreases.*

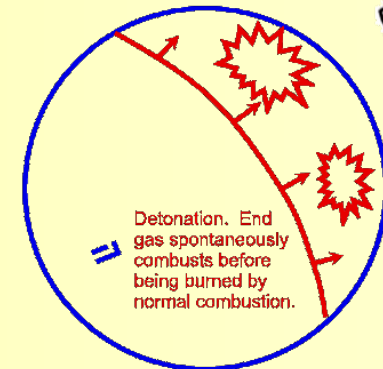
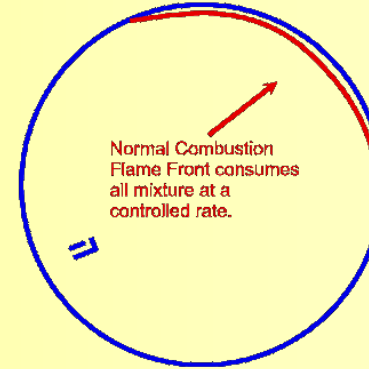
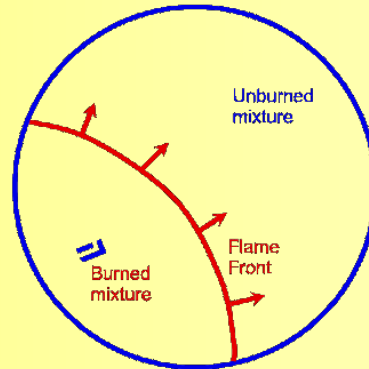
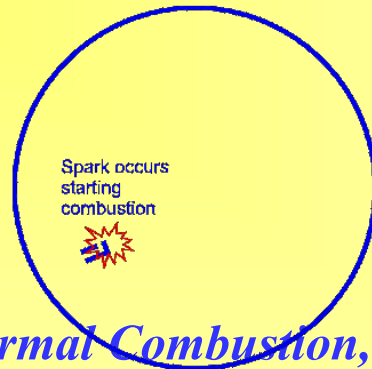


COMBUSTION IN SI ENGINES





NORMAL / ABNORMAL COMBUSTION



➤ In Normal Combustion,

- ✓ The flame initiated by spark, travels across the combustion chamber in a fairly **uniform** manner.

➤ But under Certain Operating Conditions

- ✓ **Abnormal** combustion may take place which is **detrimental** to **life** and **performance** of engine.
- ✓ This abnormal combustion includes, **detonation** or **knock**, **run-on**, **wild-ping**, **rumble** etc.
- ✓ Among the various **outcomes** of abnormal combustion,
 - **Detonation or knock is the most important phenomena because it limits on selection of compression ratio**
 - **Which governs power output and efficiency of the engine. Abnormal combustion is undesirable.**

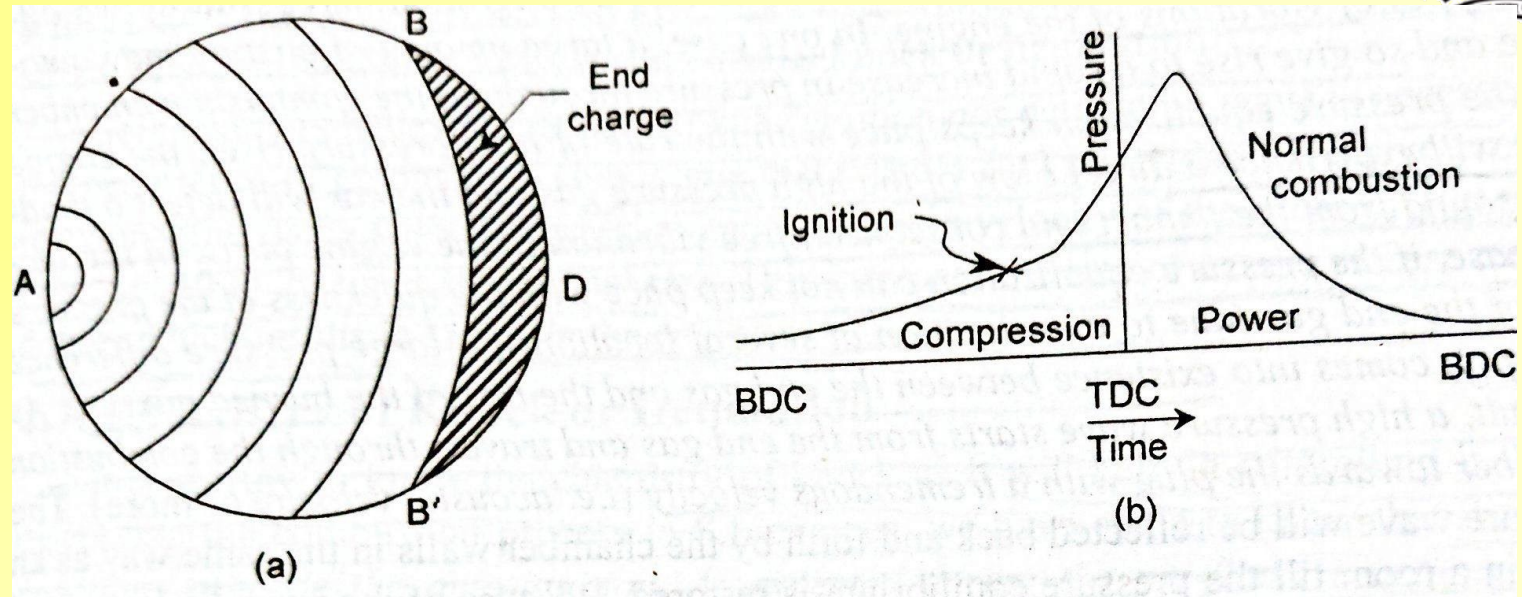
➤ Detonation or Knock

- ✓ **Detonation** or knock are the terms to signify an **usual** sharp **metallic** sound which is clearly audible under **abnormal** combustion during engine operation.
- ✓ This is very **harmful** phenomenon of combustion process, which gives a direct **blow** to engine **structure** and may lead to **failure** of the engine.
- ✓ The exact mechanism of knocking phenomenon is as yet incomplete; but analysis of high-speed photographs of combustion process and experimental data lead to **two** general theories of knock;

- ✓ (a) **Auto-ignition** theory & (b) **Wave** theory



NORMAL COMBUSTION

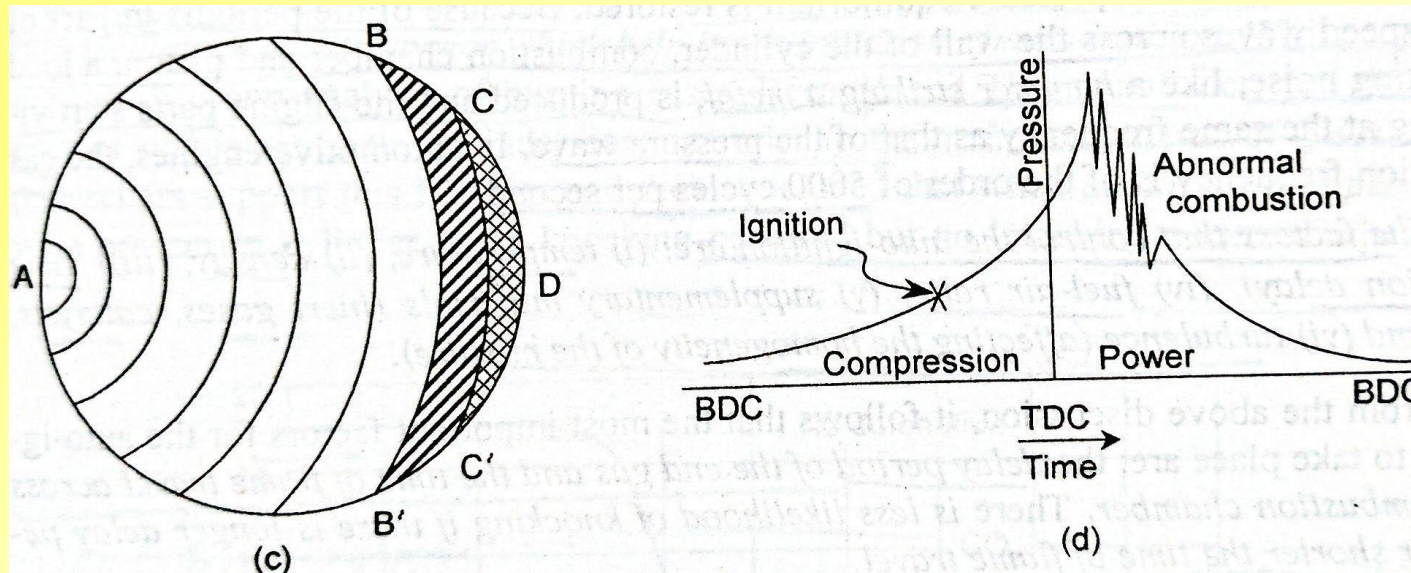


➤ Auto-ignition Theory

- ✓ After ignition, **stable** flame nucleus is framed and **propagate** molecule to molecule
- ✓ When flame reaches to **far side** of C.C. and pass through the **last part** of unburned charge, the combustion pressure is normally attained.
- ✓ Now consider the burning of **last part of charge**/end gas, which is under **compression** by
 - **Piston due to compression stroke and**
 - **Advancing the flame front.**
- ✓ The pressure of end gas when flame front is about to reach, **almost** equal to peak pressure of cycle, as by this time most of the charge (**90 to 95%**) has already been burnt.
- ✓ **Temperature** and **density** of **end gas** will increase due to
 - **Compression and Radiation heat transfer from the advancing flame.**
- ✓ Thus end gas **exists** at very **high** pressure & temperature.



ABNORMAL COMBUSTION & DETONATION



➤ Detonation...

- ✓ If the last /end part of mixture is rapidly **compressed** so that its temperature **exceeds** to the self-ignition temperature, auto-ignition of end gas may occurs **without** assistance of initiating flame.
- ✓ If **delay period** of end charge is sufficiently **long**, the flame front will reach it **before** the end of delay period and combustion of end charge (area BB'D) will be in **normal manner**, is called **Normal** combustion.
- ✓ If the ignition delay of end charge is **consumed before** the flame (CC') has arrived, auto-ignition (area CC'D) may takes place. In such condition auto-ignition occurs & combustion is **abnormal**.
- ✓ Thus knocking in SI engine is characterized by **sudden auto-ignition of charge near the end of combustion process**.

➤ Control of Autoignition: if the **last portion of the end gas** should have

- ✓ low **temp**, low **density**, **long ignition delay**, lean/rich **mixture** & **shorter flame travel time**, knock controlled.



NORMAL & DETONATION CYCLES

➤ The p - V diagram shows a

- ✓ **Normal** cycle (ABCD) and **autoignition** (BEFGH).
- ✓ The **peak** pressure of normal cycle p_c is 50 to 60 atm.
- ✓ The end charged is compressed from p_B to p_E by advancing flame front; ($p_E \sim p_C$)
- ✓ End gas raises pr from p_E to p_F (120-200 atm)

➤ There is less likelihood of knocking, if

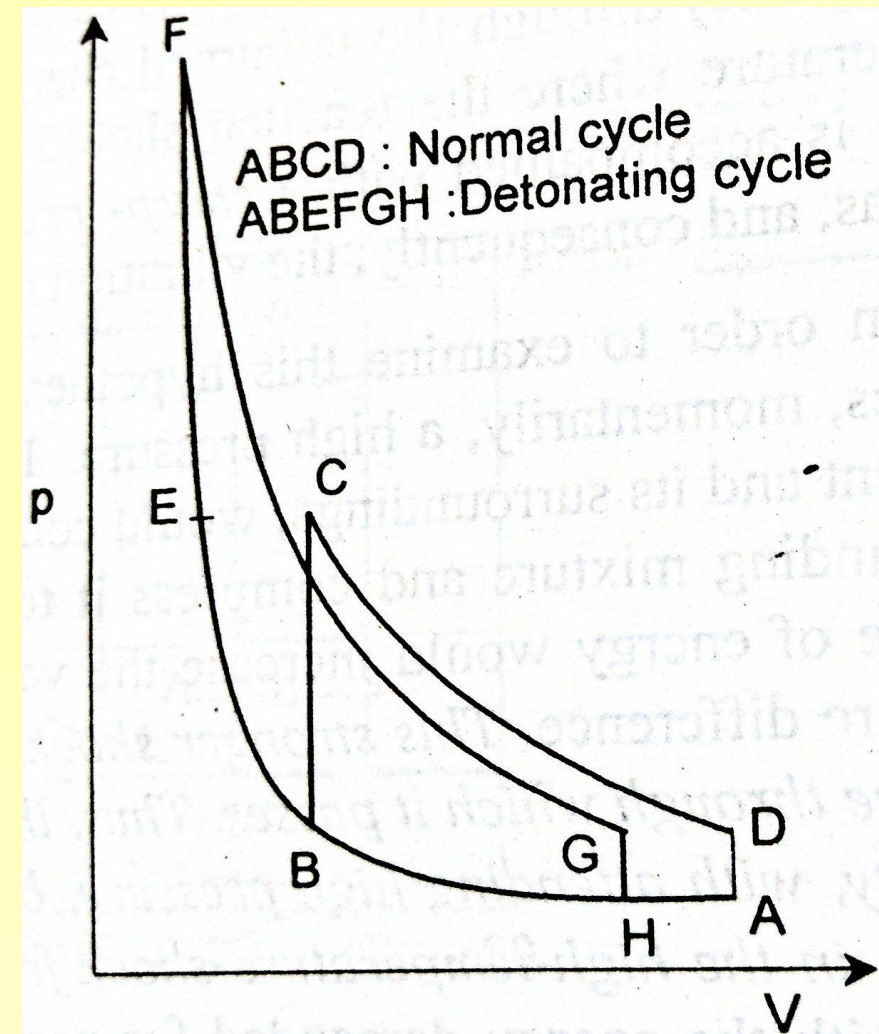
- ✓ (i) there is **longer delay** period and
- ✓ (ii) **shorter** the time of flame travel.

➤ Suggestion to avoid Detonation,

- ✓ Presence of aldehydes / ketones and peroxides must be **avoided** either by
 - **Selecting** proper fuel or
 - Add **additives/dopes** in fuels.

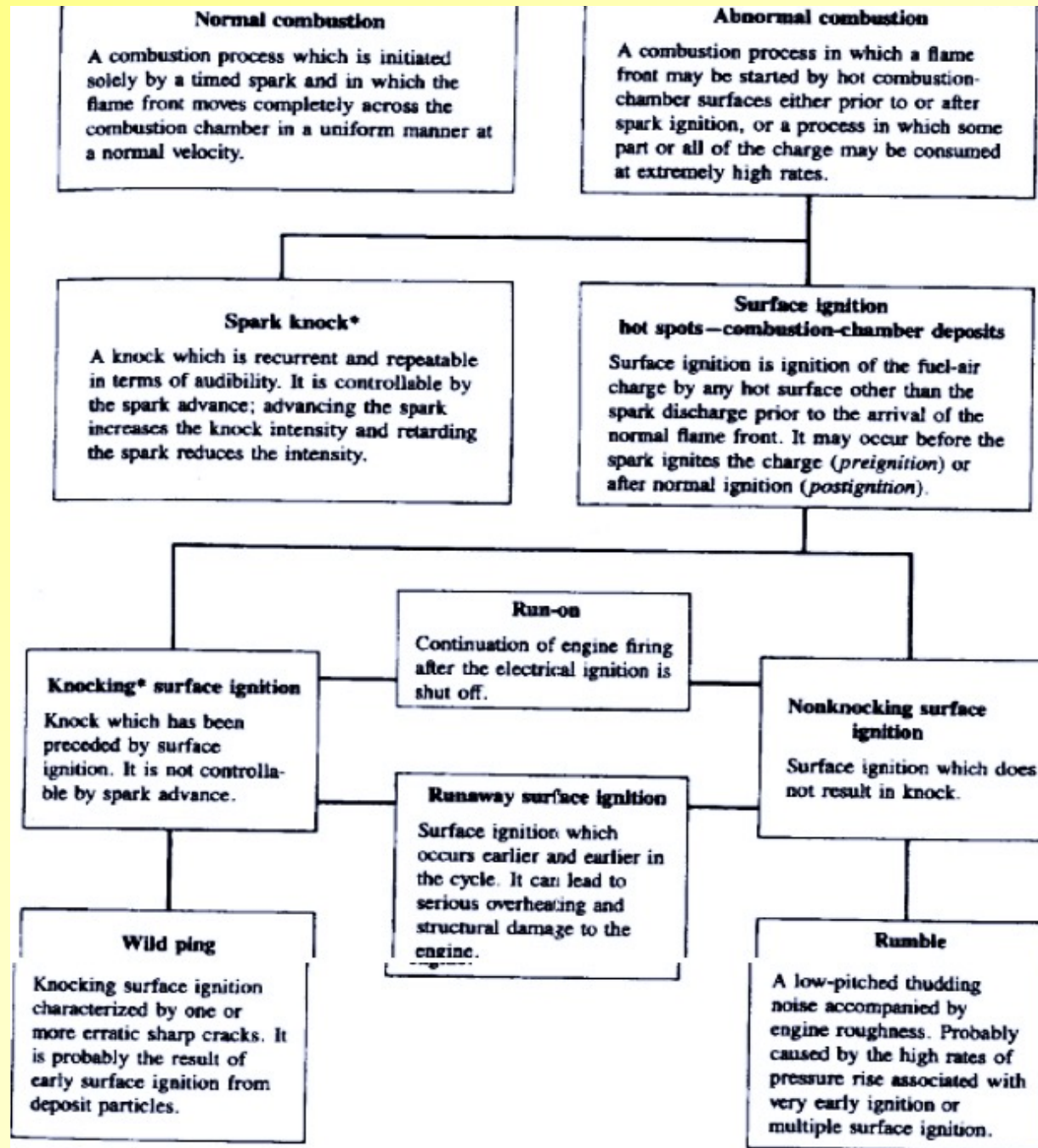
➤ Affect of knock in SI engines

- ✓ Load pulsating **noise** & mechanical **failure** of engine
- ✓ **Melting** of piston, rings, engine parts due to **overheating**
- ✓ **Pre-ignition** of fresh charge before sparking





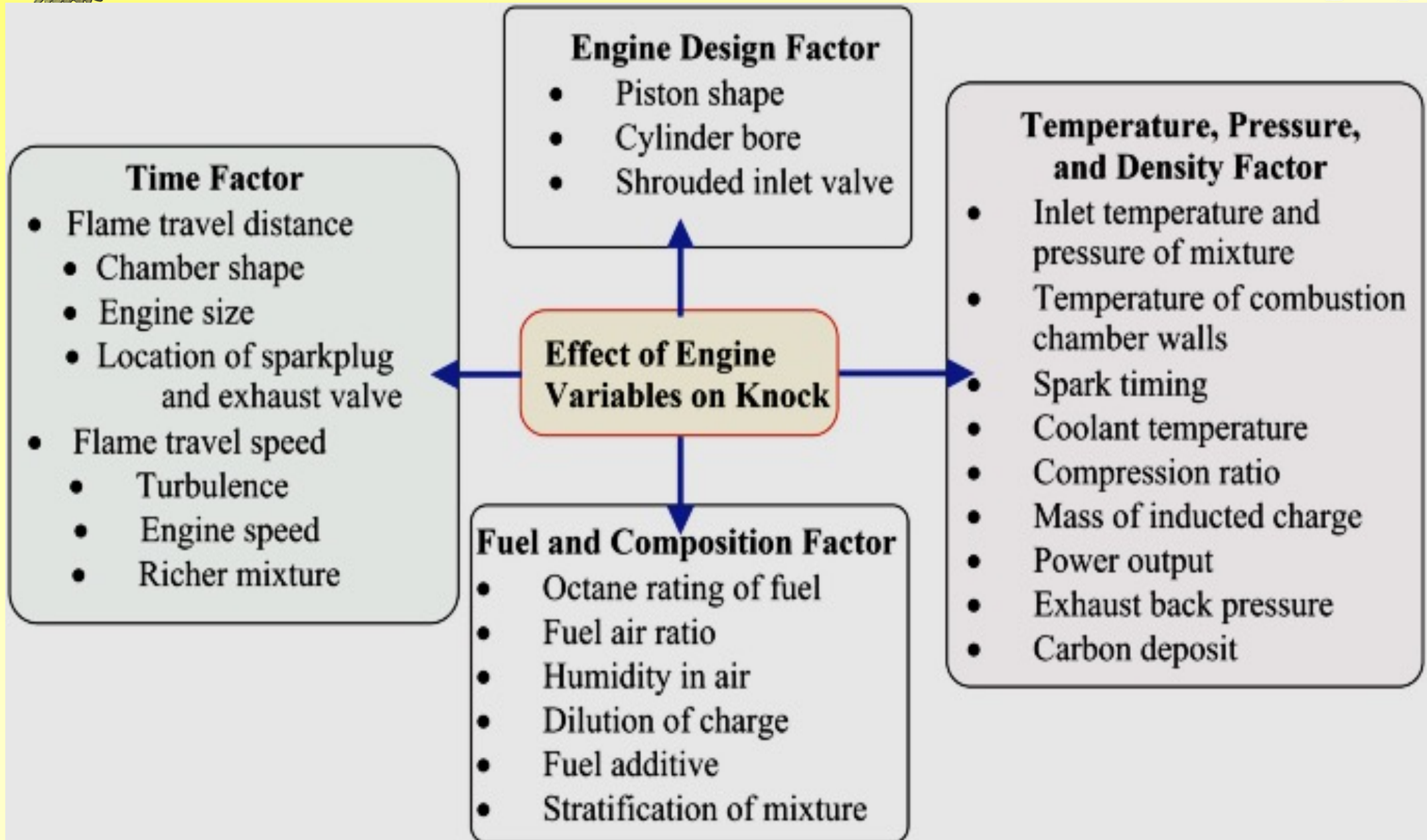
FACTORS EFFECTING KNOCK



*Knock: The noise associated with autoignition of a portion of the fuel-air mixture ahead of the advancing flame front.



EFFECT OF ENGINE VARIABLES ON KNOCK





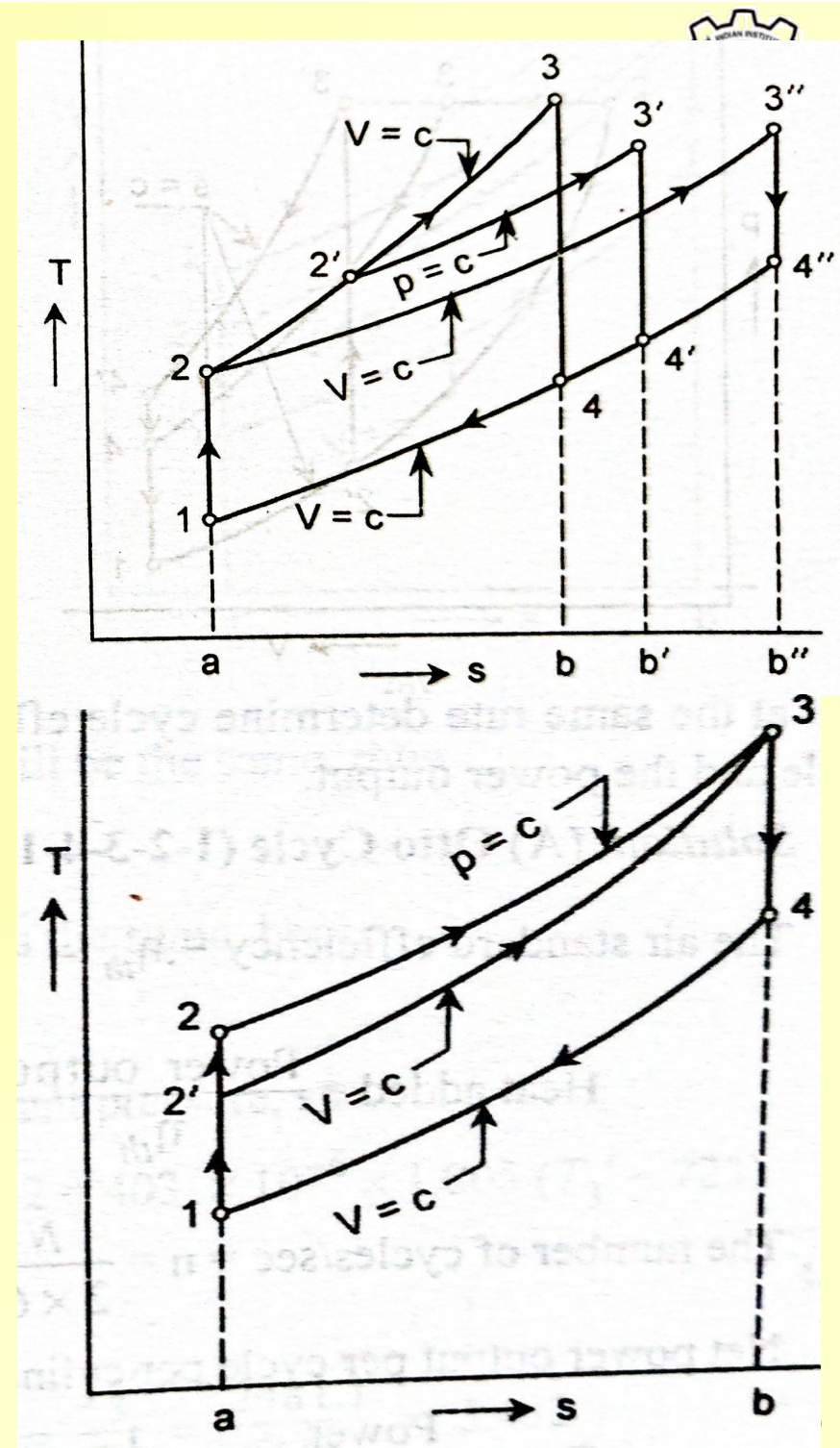
DETONATION RATING : SI FUELS

- *Octane rating is the Ability of SI fuels to measure*
 - ✓ *Resistance* to knock/detonation during combustion.
- *Octane Number: the percentage of iso-Octane (C_8H_{18}) and normal Heptane (C_7H_{16}) (by volume) in a mixture which matches the knocking tendency of fuel under standard operating conditions, is termed as Octane Number rating of the fuel.*
 - ✓ Iso-octane and normal heptane are known as primary fuels.
 - ✓ The fuels which match with primary fuels are called secondary fuels.
- *Fuel rating is accomplished by comparing its performance with standard fuel,*
 - ✓ A combination of *iso-octane* and normal *heptane*.
- *Iso-octane, being a very Good Anti-knock fuel is*
 - ✓ Assigned a rating of *100-Octane* Number.
- *Normal heptane has very Poor anti-knock qualities is*
 - ✓ Assigned a rating of *0-Octane* Number.
- *By mixing these two fuels in varying proportion (by volume)*
 - ✓ Any blends have *anti-knock* quality vary from *0 to 100* octane number.
- *Higher the octane number rating of a fuel*
 - ✓ The greater resistance to knock and
 - ✓ High compression ratio may be used without knocking.



HUCR DOPE

- To provide a standard measure of fuel's knock characteristics, a scale was devised in which fuels are assigned an octane number ON.
- The first knock scale was developed by Ricardo (1921) and fuels were rated by the **Highest Useful Compression Ratio (HUCR)** used in engine under a given set of conditions.
- The highest useful compression ratio at which a fuel can be used without detonation in a specified engine under
 - ✓ Specific operating conditions and
 - ✓ Ignition mixture strength is adjusted to give best efficiency and anti knock measure of the fuel.
- The method is useful in determination of Octane Number of fuel.
- The comparison of thermal efficiencies for Otto, Diesel and Dual cycle engines for same CR, Q_{loss} and $p-T_{max}$ are,
 - ✓ For same CR : $\eta_{th.o} > \eta_{th.diesel} > \eta_{th.dual}$
 - ✓ For Heat loss Q_r : $Q_{r.o} < Q_{r.diesel} < Q_{r.dual}$
 - ✓ For p_{max} & T_{max} : $\eta_{th.diesel} > \eta_{th.dual} > \eta_{th.o}$

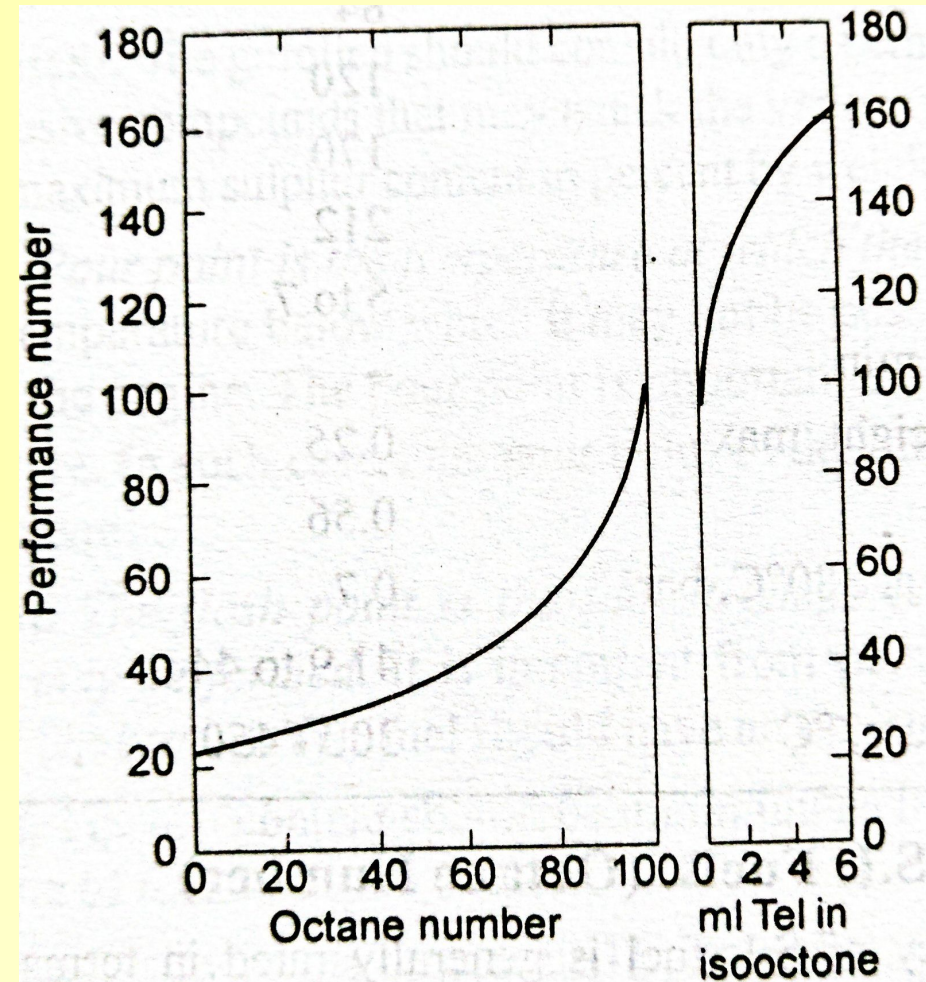




DOPE FOR SI-FUELS

- **Anti-Knock Additives**, called **Dopes** are substances
 - ✓ Added in a very **small** proportion in fuel, **improves** the knock fuel-rating and **antiknock** effect.
 - ✓ Addition of **5 ml** of **TEL** $\text{Pb}(\text{C}_2\text{H}_5)_4$ in one **Gallon** (4 L) of 80-octane petrol increases ON to **90**.
 - ✓ Other type of dopes are $\text{Fe}(\text{CO})_5$ and $\text{Ni}(\text{CO})_4$ but they are not as much effective as TEL.
 - ✓ Some **nonmetallic** substances, aniline ($\text{C}_6\text{H}_7\text{N}$), methyl aniline ($\text{C}_7\text{H}_9\text{N}$), ethyl aniline ($\text{C}_8\text{H}_{11}\text{N}$), toluidine ($\text{C}_7\text{H}_{11}\text{N}$), xylydine ($\text{C}_8\text{H}_{11}\text{N}$), phenol ($\text{C}_6\text{H}_6\text{O}$) and benzene (C_6H_6) also exhibit as dopes
- **The mechanism of TEL action is to:**
 - ✓ **Pre-cool** flame reactions, appearance of **cool flame**, **induction** period and hot **flame** propagation
- **Performance Number (PN) of fuel is based on IHP**
 - ✓ Iso-octane is assigned the PN value of **100**.
- **A supercharged engine of $\text{PN} = 130$ can produce ~ 1.3 times power without knock to 100 PN, iso-octane.**

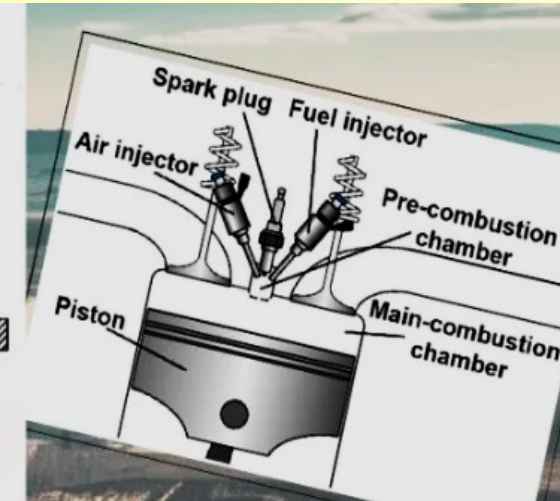
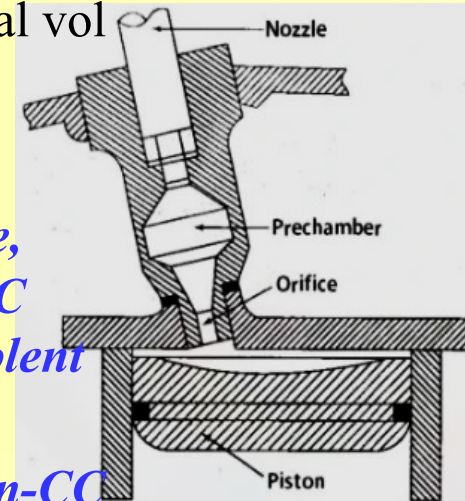
$$\text{ON} = 100 + \frac{\text{PN} - 100}{3}$$



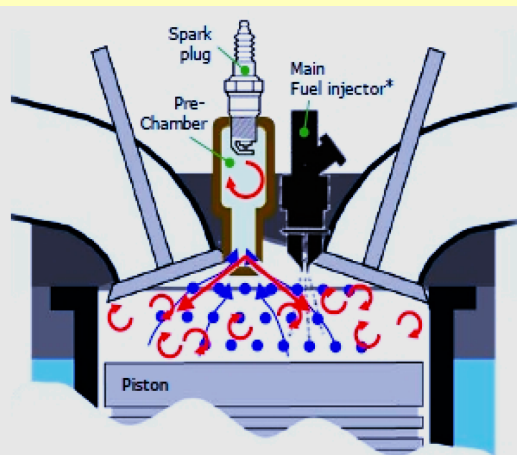


PRE-IGNITION COMBUSTION CHAMBER

- *Some engines have pre and main combustion chambers*
 - ✓ Two CCs connected each-other thru small passages
 - ✓ Pre-(ante-chamber) occupy $\sim 25\% - 35\%$ of total vol
- *At compression $\sim 1/3$ air in pre-CC & sparked it*
- *Pre-CC designed to run hot, reduces delay period.*
- *Combustion starts first in pre-CC, results in pr rise, forces the air and flaming fuel droplets in main-CC through small passages & creates a turbulence-violent air motion, avoid knocking.*
- *Violent air help in rapid mixing & burning in main-CC*
- *Creates turbulence, rapid combustion, reduces knock.*



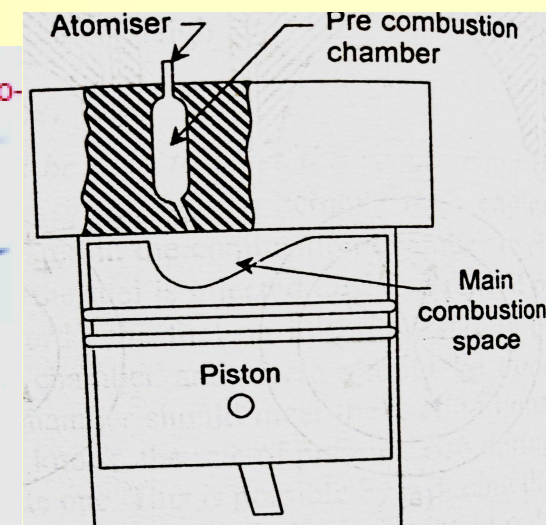
(a) Conventional S.I.



(b) Passive Pre-Chamber



(c) Active Pre-Chamber



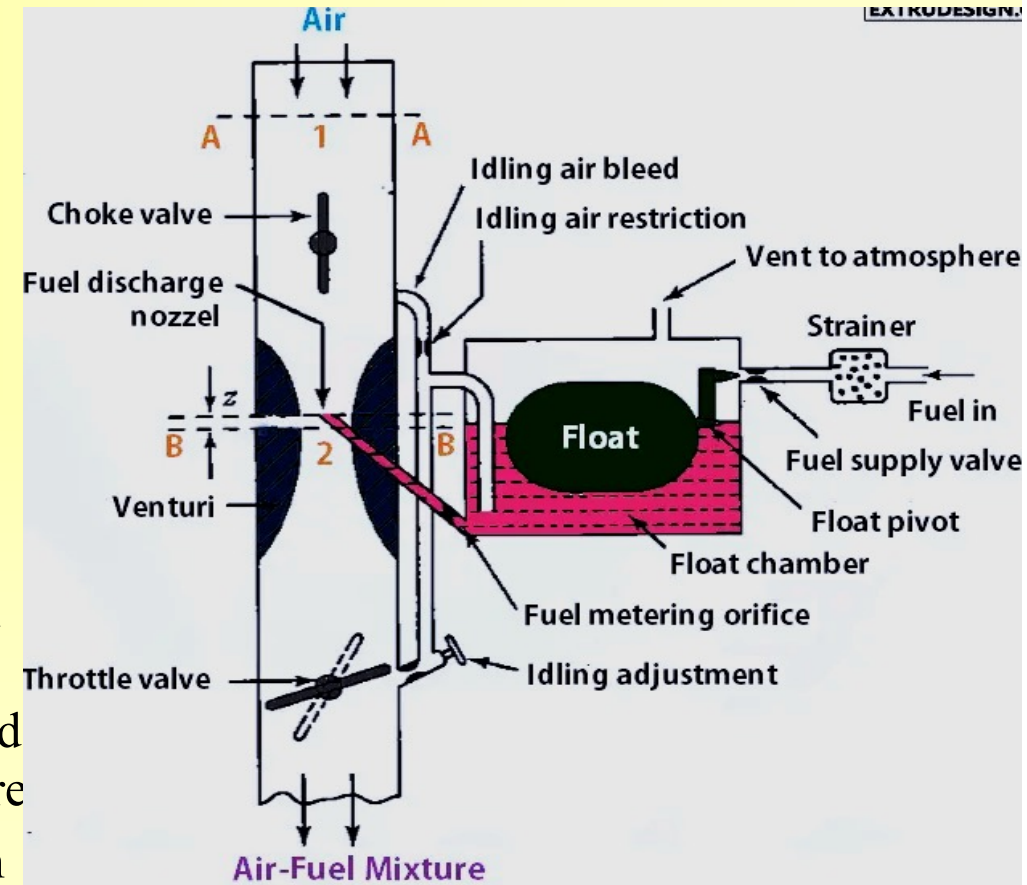


CARBURATION

➤ *Carburation is a process of formation of homogeneous combustible mixture by mixing in proper amount of fuel with air in carburettor before admission into engine cylinder.*

➤ *Mechanism of Carburation*

- ✓ Air –fuel mixture is drawn into engine cylinder through **carburettor** in **suction** stroke, due to
 - *Increasing volume of cylinder,*
 - *Decreasing pressure in combustion chamber &*
 - *Downward movement of piston.*
- ✓ **Pressure** difference between atmosphere and cylinder causes fuel-air to flow into the cylinder
- ✓ In carburettor, fuel discharged in **throat** air from a jet (**atomization**) ($\Delta p \sim 4 - 5 \text{ cm}_\text{Hg}$) due to high air **velocity** pass through the intake manifold (**vaporization**) makes a **combustible** A+F mixture
- ✓ **Amount** of fuel in high vel **air** stream depend on Δp between float chamber-jet & venturi-tube.
- ✓ The **restriction** is made in the form of venturi for proper **mixing** A+F & minimize **throttling** losses





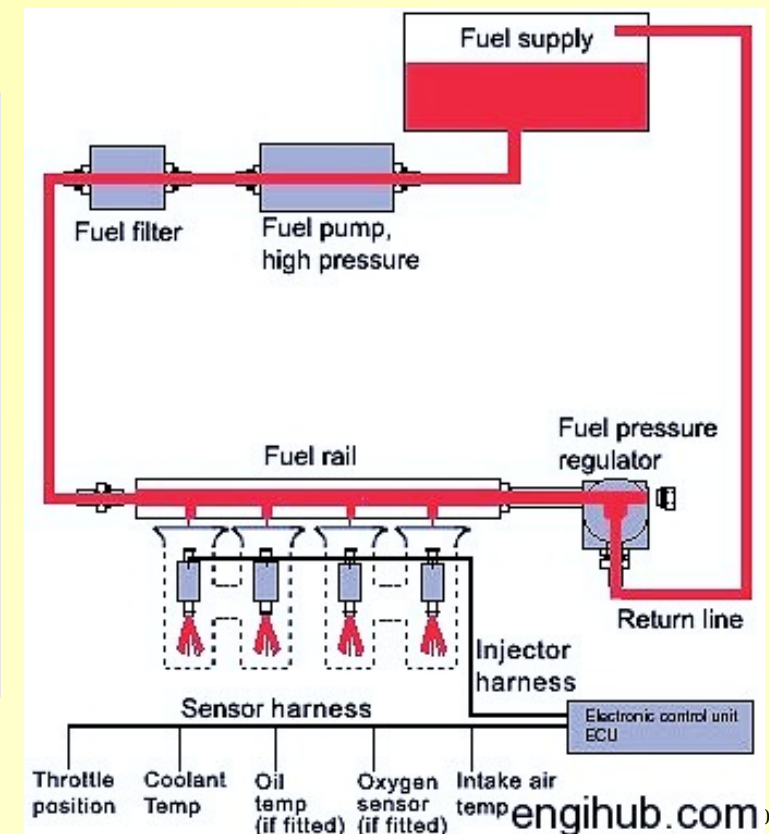
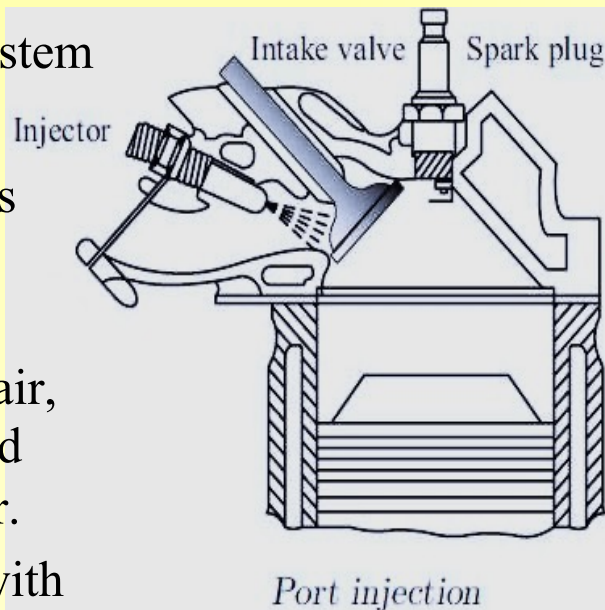
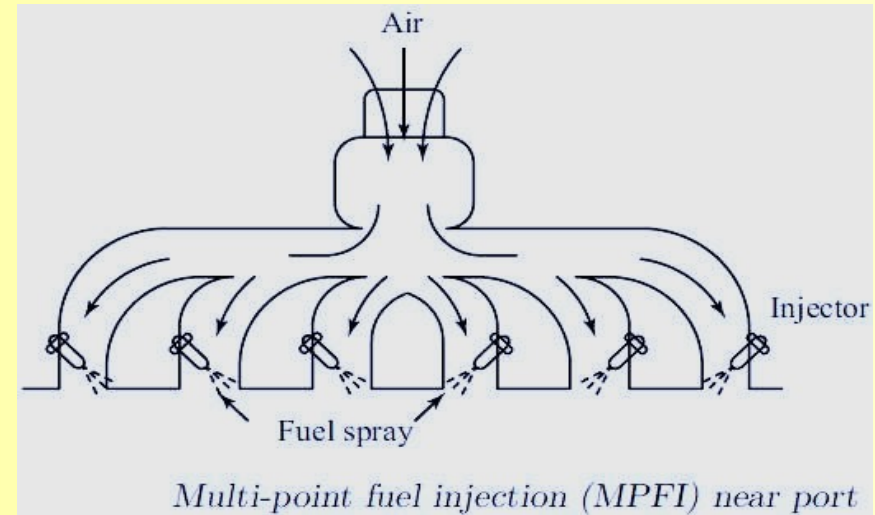
MPFI SYSTEM

- The multi-point fuel injection (MPFI) system supply air-fuel mixture in proper ratio of
 - ✓ Gasoline and Air
 - ✓ to multi-cylinder engine
- These systems function under two basic arrangements :

- ✓ 1. Port injection system
- ✓ 2. Throttle body injection system

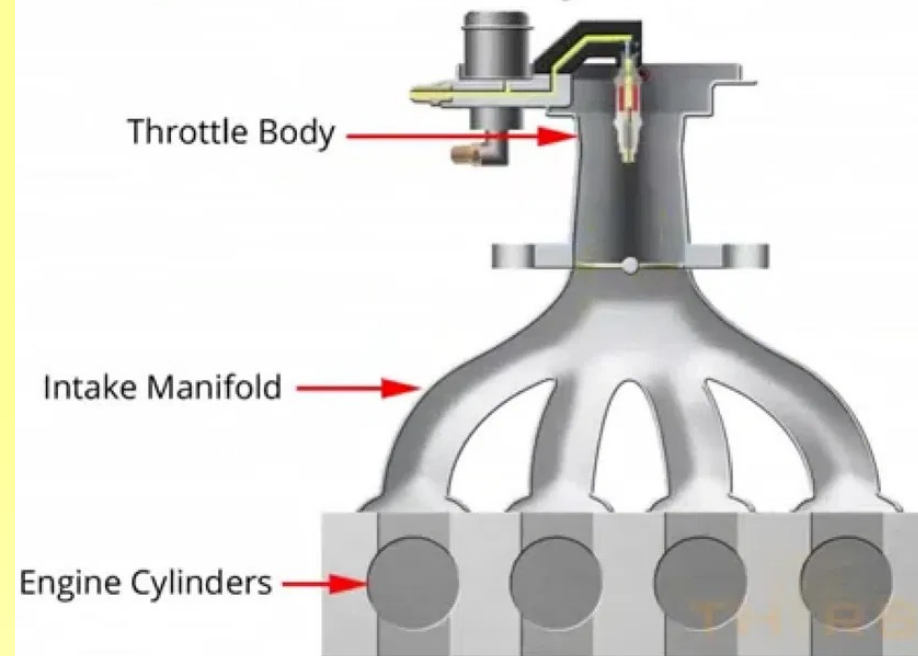
➤ Port Injection:

- ✓ In Port Injection, a injector is placed on the side of inlet **manifold** near intake port,
- ✓ Injector **spray** gasoline into air, mixes in uniform manner and goes into the engine cylinder.
- ✓ **Every** cylinder is provided with an individual injector in its intake manifold passages.

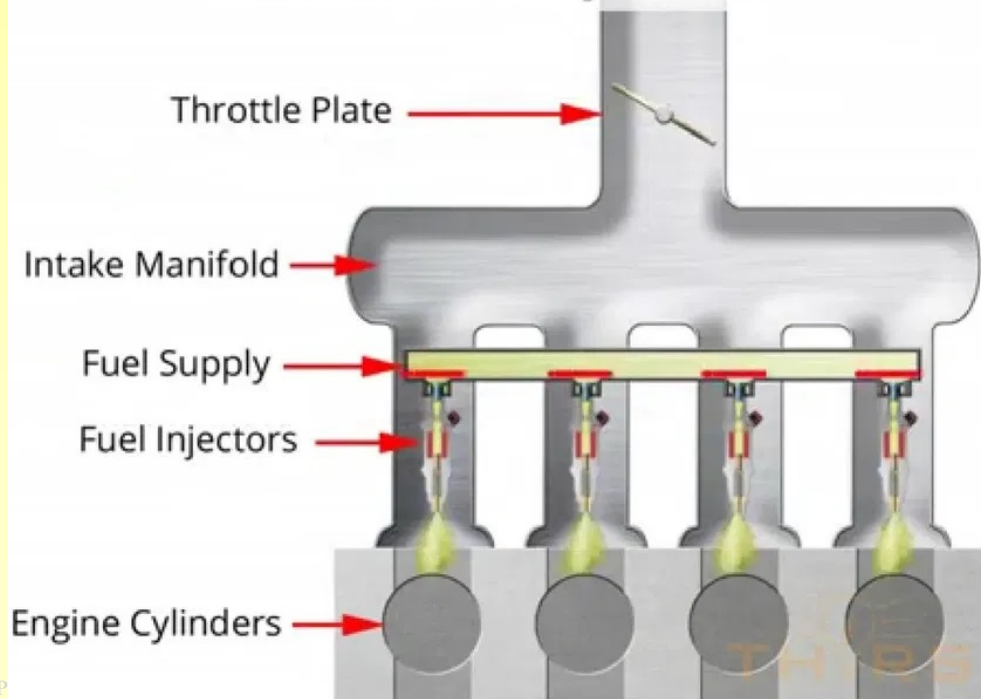




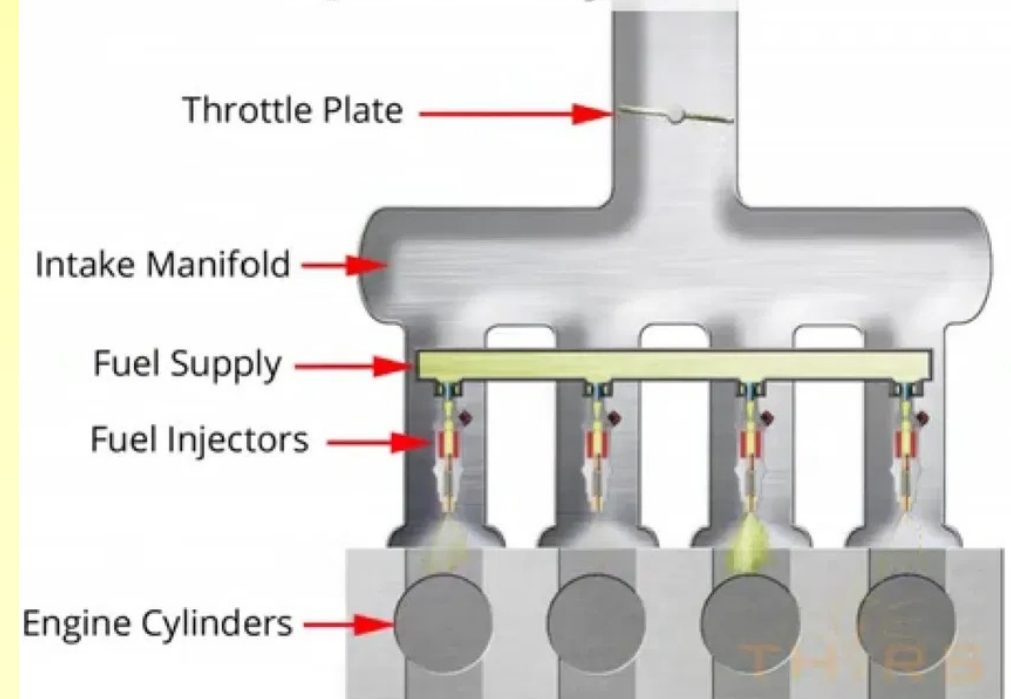
SINGLE-POINT INJECTION



MULTI-POINT INJECTION



SEQUENTIAL INJECTION





MPFI SYSTEM ...

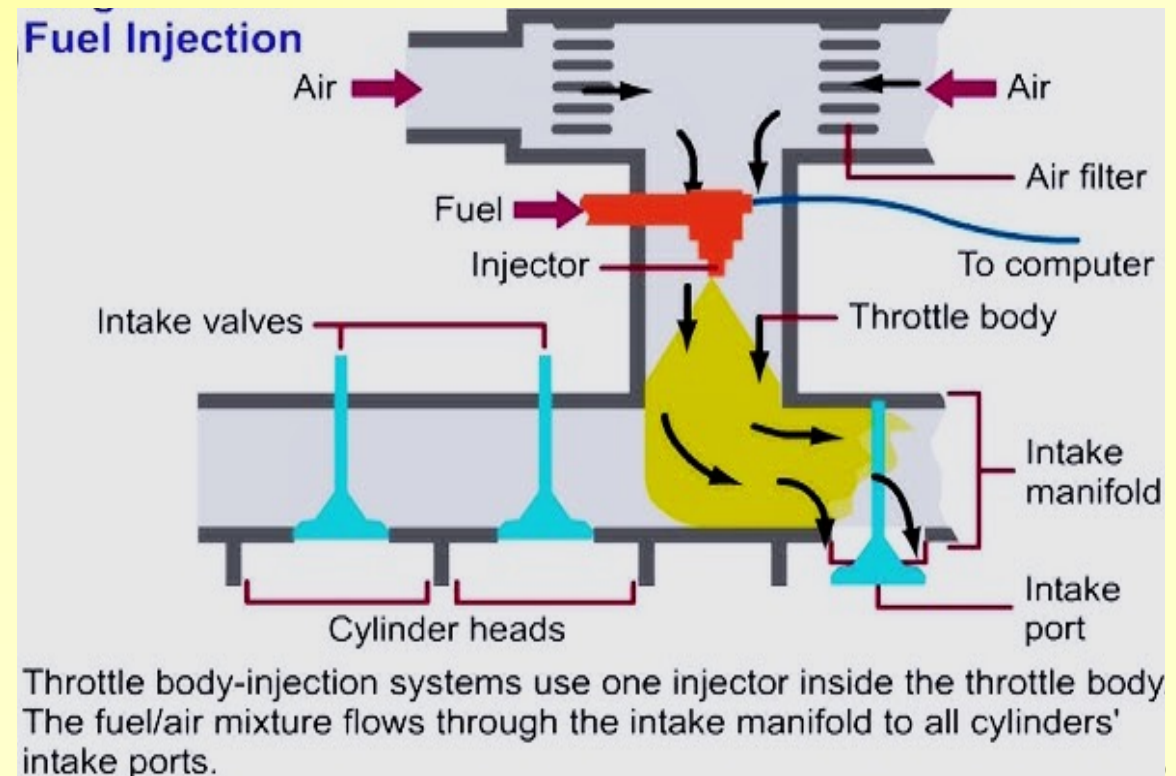
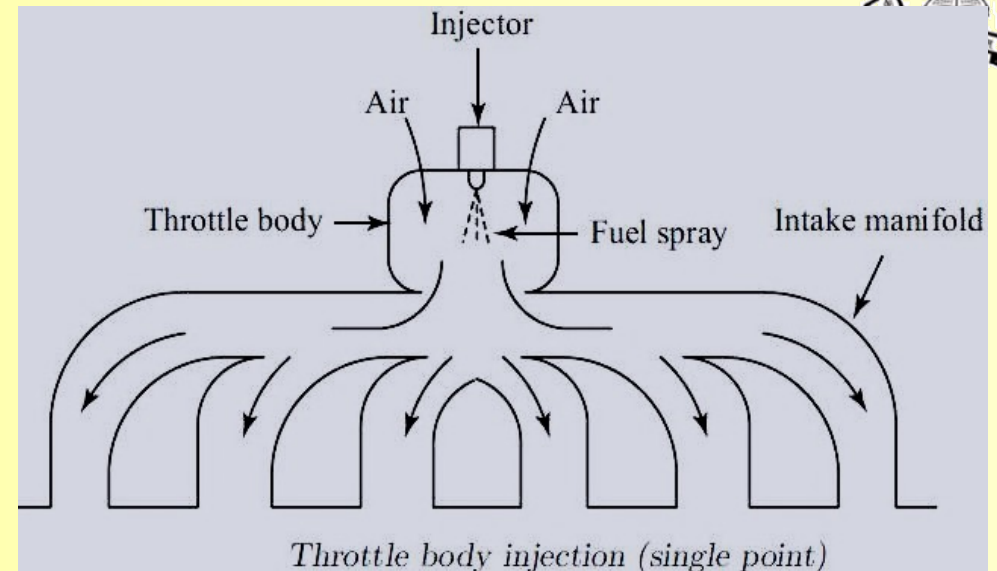


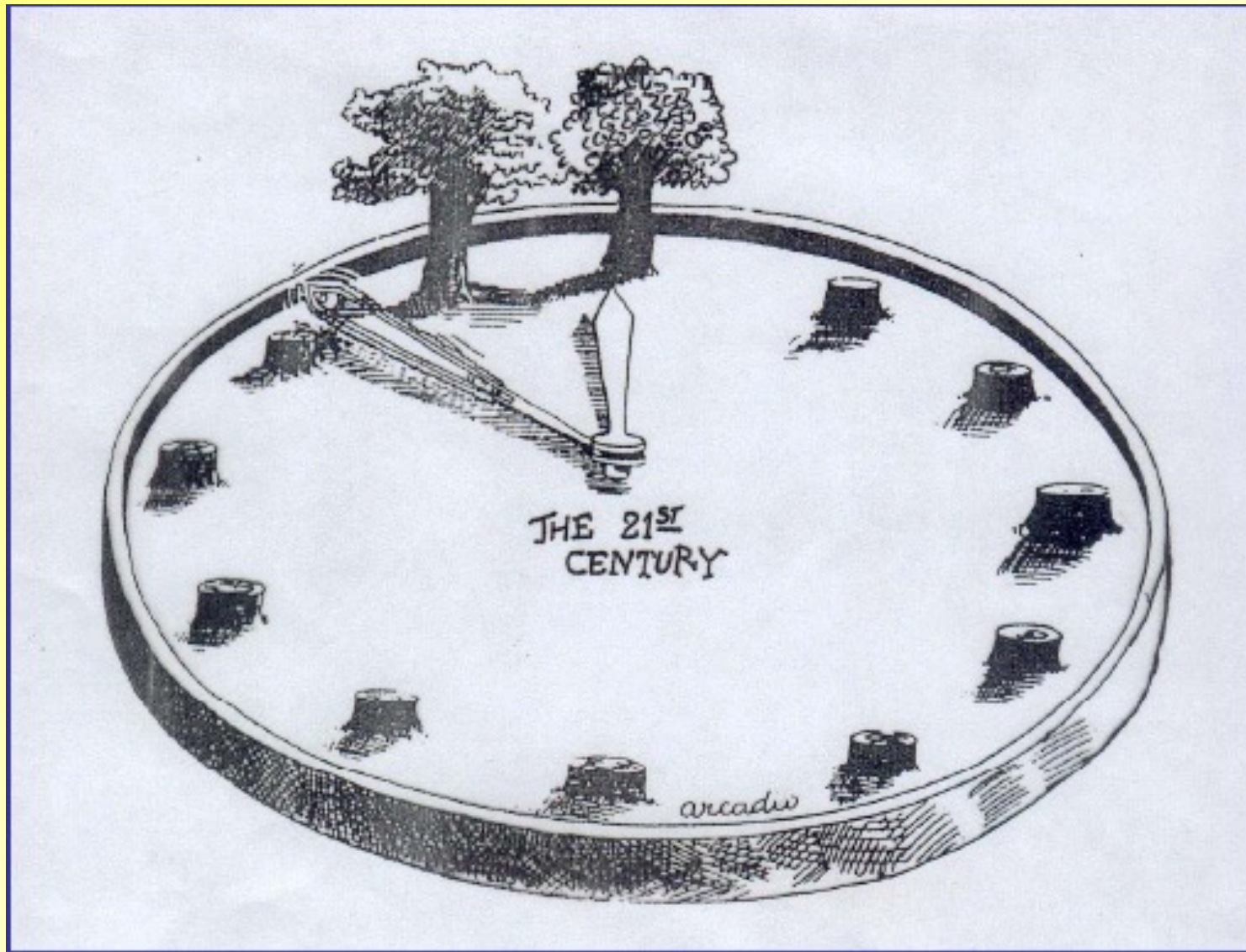
➤ *Throttle - Body Injection System:*

- ✓ Throttle body fuel-injection system is **similar** to a carburettor with a throttle valve controlling the amount of air-fuel entering into the intake manifold.
- ✓ Injector is placed slightly above the **throat** of throttle body, sprays gasoline into the air stream to intake manifold where it **mixes** with air then passes through throttle body.
- ✓ Then enters into the **respective** intake manifold of respective cylinder.

➤ *The port injection system and throttle body injection system may be either pulsed system or continuous systems.*

- ✓ In pulse fuel injection system can be either timed or continuous type.
- ✓ In timed out injection system gasoline is sprayed continuously from the injectors.





Thank You